THE COMPLETE BUSINESS KNOWLEDGE ENCYCLOPEDIA

الموسوعة الشاملة للمعرفة في عالم الأعمال

VOLUME 3

Corporate Finance, Economics and Business Mathematics

تمويل الشركات والاقتصاد ورياضيات الأعمال

Every Essential Business Concept, Term, Process, Document, Financial Instrument, and Commercial Practice You Need to Understand the Real Business World

IN THIS VOLUME

- Part IV — Corporate Finance and Financial Management
- Part V — Economics and the Macroeconomic Environment
- Part XX — Business Mathematics and Commercial Calculations

A practical reference for finance and computer-science graduates entering the business world · First edition — October 2026

About this volume

This volume is part of a ten-volume reference collection. Chapter numbers run continuously across the collection, so a reference such as “§47.3” identifies the same section in every volume, in the master index (Volume 10), and in the learning roadmap.

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Vol. 1	Foundations and the Language of Business
Vol. 2	How Businesses Work — Accounting and Financial Statements
Vol. 3	**Corporate Finance, Economics and Business Mathematics**
Vol. 4	Banking, International Trade and Treasury
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Vol. 6	Investment, Capital Markets, Commodities and Market Intelligence
Vol. 7	Law, Entrepreneurship, Management and Technology
Vol. 8	Professional Practice — Communication, Deals and Advanced Thinking
Vol. 9	Document Library, Master Reference, Exercises and Learning Roadmap
Vol. 10	Master Table of Contents, Glossary and Unified Index

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PART IV

Corporate Finance and Financial Management

تمويل الشركات والإدارة المالية

Corporate finance answers three questions every company faces: **What should we invest in?** (capital budgeting), **How should we pay for it?** (capital structure and financing), and **How do we manage day-to-day money and return value to owners?** (working capital, cash management and dividend policy). Valuation and M&A apply the same tools to whole companies.

A finance graduate has seen most formulas. This part emphasises application: realistic assumptions, the judgement involved in each input, how practitioners present results, and the mistakes that make models look precise while being wrong. Full step-by-step treatments of every formula, with Arabic explanations, are in Part XX.

LEARNING OBJECTIVES · **Learning objectives for Part IV**

- Apply time-value-of-money tools to loans, leases, payment terms and investments.
- Evaluate projects with NPV, IRR, payback and profitability index, and know when each misleads.
- Estimate the cost of capital and reason about capital structure and dividend policy.
- Manage working capital and liquidity, and choose appropriate financing.
- Build forecasts and three-statement models, and analyse sensitivity and scenarios.
- Value a company using DCF, trading comparables and precedent transactions.
- Understand the financial mechanics of acquisitions, LBOs and restructurings.

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CHAPTER 27

Time Value of Money

LEARNING OBJECTIVES · In this chapter

Present and future value, annuities, perpetuities, effective rates, loan payments and the cost of payment terms — the arithmetic beneath every financial decision.

27.1 Core relationships

FORMULA · Future and present value

$FV = PV \times (1 + r)^n$
$PV = FV \div (1 + r)^n$

r = periodic rate; n = number of periods. With m compounding periods per year at nominal annual rate i: FV = PV × (1 + i/m)^(m×t). Continuous compounding: FV = PV × e^(i×t).

WORKED EXAMPLE · Basic calculations

- \$10,000 invested at 8% for 5 years: $FV = 10{,}000 \times 1.08^5$ = **\$14,693.28**.
- \$50,000 receivable in 3 years, discount rate 10%: PV = 50,000 ÷ 1.331 = **\$37,565.74**.
- Which is better for a seller: \$100,000 now, or \$106,000 in 9 months, if its cost of funds is 9%? PV of 106,000 = 106,000 ÷ 1.09^0.75 = 106,000 ÷ 1.06677 = **\$99,365** → take \$100,000 now (ignoring credit risk, which would make the deferred option even worse).

27.2 Annuities and perpetuities

FORMULA · Annuities and perpetuities

PV of an ordinary annuity (payments at end of each period): $PV = C \times [1 - (1 + r)^{-n}] \div r$
FV of an ordinary annuity: $FV = C \times [(1 + r)^n - 1] \div r$
Annuity due (payments at start): multiply the ordinary result by (1 + r)
Perpetuity: $PV = C \div r$
Growing perpetuity (first payment C_1, growth g < r): $PV = C_1 \div (r - g)$
Loan payment (annuity): $PMT = P \times r \div [1 - (1 + r)^{-n}]$

WORKED EXAMPLE · Annuities in business

- A lease of \$1,000 per year for 5 years at 8%: PV = 1,000 × 3.99271 = **\$3,992.71**.
- A business loan of \$200,000 over 5 years, monthly payments, 9% nominal annual rate: r = 0.75% per month, n = 60. PMT = 200,000 × 0.0075 ÷ (1 − 1.0075^{-60}) = 1,500 ÷ 0.36130 = **\$4,151.67 per month**. Total paid \$249,100; interest \$49,100.
- A property yielding \$100,000 a year indefinitely, required return 8%: PV = **\$1.25m**; if rents grow 3% a year: 100,000 ÷ (0.08 − 0.03) = **\$2.0m**. Growth assumptions dominate value — the same logic drives terminal values in DCF.

27.3 Nominal, effective and real rates

FORMULA · Effective annual rate (EAR) and real rate

EAR = (1 + i/m)^m − 1
Fisher relation: (1 + nominal) = (1 + real) × (1 + inflation)

- A loan at "12% per annum, monthly" has EAR = $1.01^{12} - 1$ = **12.68%**.
- A deposit at 6% with inflation of 4%: real return = 1.06 ÷ 1.04 − 1 = **1.92%**.
- Banks may quote "flat" rates on consumer and some SME loans (interest on the original principal for the whole term), which roughly double the effective rate on an amortising loan; always compute the effective rate.

COMMON MISTAKES · Day-count conventions

Interest for a period = principal × rate × (days ÷ day basis). USD money-market and SOFR loans normally use actual/360; GBP actual/365; bonds often 30/360. On $10m for 90 days at 6%: actual/360 gives $150,000; actual/365 gives $147,945. Know which convention your facility agreement specifies.

27.4 The cost of payment terms

Credit terms are a price. Offering a customer 90 days instead of 30 is a hidden discount; a supplier's early-payment discount is an investment opportunity.

WORKED EXAMPLE · Should we take the 2/10 net 30 discount?

Paying on day 10 instead of day 30 earns a 2% discount for paying 20 days early. - Simple annualised return = (2 ÷ 98) × (365 ÷ 20) = **37.2%**. - Effective annual return = (1 + 2/98)^(365/20) − 1 = **44.6%**. If the company can borrow at 9% on its overdraft, taking the discount is extremely profitable. Many companies forgo it because of poor cash planning — an expensive habit.

27.5 Spreadsheet and code tools

Purpose	Excel / Google Sheets	Python (numpy-financial)
Present value	`=PV(rate, nper, pmt, [fv], [type])`	`npf.pv(rate, nper, pmt, fv)`
Payment	`=PMT(rate, nper, pv)`	`npf.pmt(rate, nper, pv)`
NPV (cash flows from period 1)	`=NPV(rate, CF1:CFn) + CF0`	`npf.npv(rate, [CF0, CF1, ...])` (CF0 at t=0)
IRR	`=IRR(CF0:CFn)`	`npf.irr([CF0, ...])`
Irregular dates	`=XNPV(rate, values, dates)`, `=XIRR(values, dates)`	custom function or `scipy.optimize`
Effective rate	`=EFFECT(nominal, m)`	`(1 + i/m)**m - 1`

COMMON MISTAKES · Spreadsheet trap

Excel's `NPV` function assumes the first cash flow occurs one period from now. Writing `=NPV(10%, CF0:CF5)` discounts the initial investment by one year and overstates NPV. Use `=NPV(10%, CF1:CF5) + CF0`.

PRACTICE · Practice questions

1. Compute the monthly payment on a $500,000, 7-year loan at 8% nominal annual, monthly payments.
2. A customer offers to pay $1,030,000 in 120 days instead of $1,000,000 in 30 days. Your cost of funds is 10%. Which is better (ignore credit risk)?
3. Convert a nominal 10% compounded quarterly into an EAR.
4. A supplier offers 1/15 net 60. What is the effective annual cost of not taking the discount?

ANSWER · Answers to the practice questions

1. r = 0.08 ÷ 12 = 0.006667; n = 84; $(1.006667)^{84}$ = 1.7468; PMT = 500,000 × 0.006667 ÷ (1 − 1/1.7468) = 3,333.33 ÷ 0.42753 = **≈ $7,793**.
2. Compare at day 30: PV of 1,030,000 discounted 90 days = 1,030,000 ÷ (1 + 0.10 × 90/365) = 1,030,000 ÷ 1.02466 = **$1,005,210** > $1,000,000 → the deferred offer is slightly better before credit risk; with any material credit risk, the difference is too small to justify it.
3. $(1 + 0.025)^4 - 1$ = **10.38%**.
4. Paying 45 days early earns 1/99: EAR = (1 + 1/99)^(365/45) − 1 = 1.0101^8.111 − 1 ≈ **8.5%**. If your borrowing cost is below 8.5%, take the discount; if above, pay on day 60.

CHAPTER 28

Capital Budgeting: NPV, IRR and Investment Decisions

LEARNING OBJECTIVES · In this chapter

How companies decide whether to invest: estimating incremental cash flows, applying NPV, IRR, payback and profitability index, and handling the practical complications — inflation, taxes, working capital, terminal values, capital rationing and strategic options.

28.1 The decision rule

A project creates value if the present value of its expected incremental cash flows exceeds the investment — **NPV > 0** — when discounted at a rate reflecting the project's risk.

FORMULA · Net present value

$NPV = \Sigma\ (CF_t \div (1 + r)^t)$ for $t = 0 \ldots n$
where CF_0 is normally negative (investment).

IRR is the rate r* at which NPV = 0. Profitability index PI = PV of future cash flows ÷ initial investment.

28.2 Estimating incremental cash flows

Include:

- **Incremental** revenue and costs (what changes because of the project).
- **Taxes** on incremental operating profit, including the tax shield of depreciation.
- **Capex**, including installation and commissioning.
- **Working capital** investment at start and as the project grows, and its **recovery** at the end.
- **Terminal / salvage value** after tax.
- **Opportunity costs** (e.g., land the company already owns that could be sold).
- **Side effects:** cannibalisation of existing sales, or synergies.

Exclude:

- **Sunk costs** (feasibility studies already paid).
- **Allocated overheads** that do not change.
- **Financing costs** (interest) — they are reflected in the discount rate, not in cash flows (when using WACC).

FORMULA · Project free cash flow

FCF = (Revenue − Operating costs − Depreciation) × (1 − t) + Depreciation − Capex − ΔWorking capital
= EBIT × (1 − t) + D&A − Capex − ΔNWC

28.3 Worked example: a cold-storage warehouse (fictional)

A food distributor considers building a cold-storage warehouse.

Assumption	Value
Construction and equipment (t = 0)	$2,000,000
Working capital (t = 0, recovered in year 5)	$200,000
Incremental EBITDA (rent savings + third-party storage income – operating costs)	$650,000 per year, years 1–5
Depreciation	Straight-line over 5 years to zero: $400,000 per year
Tax rate	20%
Sale value of assets at end of year 5	$300,000 (fully taxable gain, as tax value is zero)
Discount rate (project WACC)	11%

Annual cash flow, years 1–5: EBIT = 650 − 400 = 250; tax 50; NOPAT 200; + depreciation 400 → **600** ($000). **Year 5 extras:** after-tax sale proceeds 300 − 60 = 240; working-capital recovery 200 → year-5 total **1,040**.

Year	0	1	2	3	4	5
Cash flow ($000)	(2,200)	600	600	600	600	1,040
Discount factor 11%	1.0000	0.9009	0.8116	0.7312	0.6587	0.5935
Present value	(2,200.0)	540.5	487.0	438.7	395.2	617.2
Cumulative cash flow	(2,200)	(1,600)	(1,000)	(400)	200	1,240

- **NPV** = **+$278.7k** → accept on financial grounds.
- **IRR ≈ 15.5%** (NPV at 15% = +30.1; at 16% = −25.9; interpolation).
- **Payback** = **3.67 years** (3 + 400/600).
- **PI** = **2,478.7 ÷ 2,200** = **1.13.**

PROFESSIONAL INSIGHT · How to present this to an investment committee

"The project requires $2.2m including working capital, generates an NPV of $0.28m at our 11% hurdle and an IRR of about 15.5%, with payback in under four years. The result is most sensitive to occupancy of the third-party storage space: if third-party income is 30% lower than planned, NPV falls to roughly zero. We recommend approval with a pre-letting target of 50% of third-party space before construction starts."

Decision-makers want the answer, the key risk and the mitigating action — not the spreadsheet.

28.4 NPV vs IRR: when they disagree

IRR is intuitive but has flaws:

1. **Scale:** a small project with high IRR can create less value than a large project with lower IRR.
2. **Timing:** IRR implicitly assumes reinvestment at the IRR.
3. **Multiple IRRs:** cash flows that change sign more than once (e.g., large decommissioning costs at the end) can produce several IRRs or none.
4. **Financing-type flows:** for a cash inflow followed by outflows (a loan), a higher IRR is worse.

WORKED EXAMPLE · Mutually exclusive projects

	Year 0	Years 1–4	IRR	NPV at 10%
Project A	(1,000)	+1,300 in year 1 only	30.0%	+181.8

	Year 0	Years 1–4	IRR	NPV at 10%
Project B	(5,000)	+1,800 per year	≈ 16.4%	+705.8

If only one can be chosen, B creates more value despite its lower IRR. Use NPV for ranking mutually exclusive projects; use IRR as a supplementary communication tool. The **modified IRR (MIRR)** fixes the reinvestment assumption by compounding inflows at the cost of capital.

28.5 Other criteria

Criterion	Strengths	Weaknesses	Typical use
NPV	Theoretically correct; additive	Requires discount rate; less intuitive	Primary decision rule
IRR	Intuitive %; no need to state rate upfront	Scale, timing, multiple IRR problems	Private equity, project finance, communication
Payback	Simple; liquidity/risk proxy	Ignores time value and post-payback cash	Screening; uncertain environments
Discounted payback	Adds time value	Still ignores later cash flows	Risky markets
PI	Useful under capital rationing	Same as NPV for single decisions	Ranking when capital is limited
Accounting rate of return	Uses reported profits	Ignores cash and timing	Legacy practice; avoid

28.6 Practical complications

- **Inflation:** be consistent — nominal cash flows with nominal discount rates, or real with real.
- **Different lives:** compare projects using equivalent annual annuity (EAA = NPV ÷ annuity factor) or a common horizon.
- **Capital rationing:** rank by PI and choose the combination maximising total NPV within the budget.
- **Risk:** adjust the discount rate for project risk (not just company WACC); or use scenario-weighted cash flows; avoid double-counting risk in both.
- **Real options:** the option to expand, delay or abandon has value that static NPV ignores — e.g., building a warehouse with foundations for a second floor.
- **Strategic projects:** "strategic" is not a substitute for analysis; quantify the strategic benefit or at least the cost of not investing.

COMMON MISTAKES · Common mistakes in capital budgeting

- Including interest in project cash flows and also discounting at WACC (double counting financing).
- Forgetting working capital or its recovery.
- Using company WACC for a project in a riskier country or business.
- Optimistic terminal values that account for most of the NPV.
- Ignoring cannibalisation of existing products.
- Treating sunk costs as relevant ("we've already spent $1m, we can't stop now").

PRACTICE · Practice questions

1. A machine costs $300,000, saves $90,000 a year before tax for 5 years, is depreciated straight-line to zero, tax 25%, discount rate 10%. Compute NPV and payback.

2. Two machines: A costs $50,000, lasts 3 years, NPV $12,000; B costs $80,000, lasts 6 years, NPV $18,000 (both at 10%). Which is better on an EAA basis?
3. Your division has a $10m budget. Projects (investment, NPV): P1 (4, 1.2), P2 (3, 1.0), P3 (5, 1.4), P4 (2, 0.5). Which combination maximises NPV?

ANSWER · Answers to the practice questions

1. Depreciation 60,000; EBIT 30,000; tax 7,500; NOPAT 22,500; cash flow = 22,500 + 60,000 = **$82,500** per year. PV = 82,500 × 3.79079 = 312,740 → **NPV = +$12,740**. Payback = 300,000 ÷ 82,500 = **3.64 years**.
2. Annuity factors at 10%: 3 years 2.48685; 6 years 4.35526. EAA A = 12,000 ÷ 2.48685 = **$4,825**; EAA B = 18,000 ÷ 4.35526 = **$4,133 → A** is better (if it can be replaced on similar terms).
3. PIs: P1 1.30 (1.2/4 = 0.30 NPV per $), P2 1.33, P3 1.28, P4 1.25. Feasible combinations within 10: P1+P2+P4 (9) → NPV 2.7; P2+P3+P4 (10) → 2.9; P1+P3 (9) → 2.6; P1+P2 (7)… → best: **P2 + P3 + P4 = $2.9m**. (Ranking by PI alone picks P2, P1, then cannot fit P3; checking combinations matters when projects are indivisible.)

CHAPTER 29

Cost of Capital, Capital Structure and Dividend Policy

LEARNING OBJECTIVES · In this chapter

How to estimate the cost of equity, the cost of debt and WACC; how companies choose between debt and equity; and how they decide how much cash to return to shareholders.

29.1 The cost of capital

The cost of capital is the return investors require for providing funds to an investment of a given risk — the opportunity cost of their money. It is a **market-based, forward-looking** concept, not the interest rate a company happens to pay.

Cost of equity

FORMULA · CAPM

Ke = Rf + β × ERP (+ country risk premium + size or specific premiums where justified)

Rf = risk-free rate (government bond yield in the currency of the cash flows); β = sensitivity of the stock to the market; ERP = equity risk premium (expected market return above Rf).

WORKED EXAMPLE · Estimating Ke for a fictional Gulf-based logistics company (USD cash flows)

- Rf: 10-year US Treasury yield, assume 4.5%.
- Beta: relevered peer beta 1.2.
- ERP: assume 5.5% (practitioners commonly use 4–6% for mature markets; sources such as academic surveys and published estimates vary).
- Country risk premium: assume 1.5–3% depending on country, e.g., 2.0%.
- Ke = 4.5% + 1.2 × 5.5% + 2.0% = **13.1%**.

Alternatives and cross-checks: dividend discount model (Ke = $D_1 \div P_0 + g$); build-up methods for private companies; investors' actual target returns (private equity often 20%+ for equity in emerging markets).

Beta: levering and unlevering

Peer betas reflect peers' leverage. To use them, unlever to an asset beta and relever at the target's capital structure:

FORMULA · Hamada (no debt beta, constant debt)

$\beta L = \beta U \times [1 + (1 - t) \times D/E]$
$\beta U = \beta L \div [1 + (1 - t) \times D/E]$

Cost of debt

The **pre-tax cost of debt** is the yield the company would pay on new long-term borrowing today (market yield on its bonds, or reference rate + credit margin for loans). Interest is usually tax-deductible, so the after-tax cost is Kd × (1 − t). Use the marginal rate of new debt, not the historical coupon.

WACC

FORMULA · Weighted average cost of capital

WACC = [E ÷ (D + E)] × Ke + [D ÷ (D + E)] × Kd × (1 − t)
Use target or market-value weights, not book values.

Continuing the example: Kd 7% pre-tax, tax 15% → 5.95% after tax; target weights E 65%, D 35% → WACC = 0.65 × 13.1% + 0.35 × 5.95% = 8.52% + 2.08% = **10.6%**.

COMMON MISTAKES · Common mistakes with WACC

- Using book weights, or the current (temporary) capital structure instead of the target.
- Mixing currencies: a USD risk-free rate with local-currency cash flows.
- Using the company WACC for every project regardless of risk.
- Spurious precision: WACC estimates have a range of ±1–2 percentage points; show sensitivity.

29.2 Capital structure: debt or equity?

	Debt	Equity
Cost	Lower (priority claim; tax-deductible interest)	Higher (residual claim)
Obligation	Fixed payments; default if missed	No fixed obligation
Control	Covenants, but no ownership	Dilution of ownership and control
Risk to company	Raises financial distress risk	Reduces it
Upside	Lenders get only interest	Shareholders get all residual upside
Availability	Requires cash flow, collateral, track record	Requires investors who believe in growth
Signalling	Confidence in cash flows	May signal shares are overvalued (for listed firms)

Leverage magnifies returns — both ways

WORKED EXAMPLE · Effect of leverage on ROE (fictional)

Assets $10m; tax 20%; debt cost 8%.

Scenario	EBIT	Unlevered ROE (equity 10m)	50% debt ROE (equity 5m; interest 0.4m)
Good year	1.5m	1.2 ÷ 10 = **12.0%**	(1.5 – 0.4) × 0.8 ÷ 5 = **17.6%**
Bad year	0.5m	0.4 ÷ 10 = **4.0%**	(0.5 – 0.4) × 0.8 ÷ 5 = **1.6%**
Very bad year	0.2m	0.16 ÷ 10 = **1.6%**	(0.2 – 0.4) × 0.8 ÷ 5 = **–3.2%** (and interest cover 0.5×)

Leverage raises returns when return on assets exceeds the cost of debt and lowers them — sharply — when it does not.

Theories in one paragraph each

- **Modigliani–Miller (1958):** in perfect markets without taxes, capital structure does not affect firm value; the cost of equity rises with leverage to offset cheaper debt. With corporate taxes, debt adds value through the interest tax shield.
- **Trade-off theory:** firms balance the tax benefits of debt against the expected costs of financial distress (lost customers, fire sales, management distraction, legal costs). Stable, asset-rich, profitable firms can carry more debt.

- **Pecking order:** because managers know more than investors, firms prefer internal funds, then debt, then equity as a last resort.
- **Practice:** CFOs target a credit rating or leverage range (e.g., net debt/EBITDA 1.5–2.5×), keep liquidity headroom, match debt maturity and currency to assets, and preserve flexibility for opportunities.

Hybrid and alternative instruments

Subordinated debt, mezzanine finance, convertible bonds, preference shares, shareholder loans, sukuk (Sharia-compliant certificates) and leases sit between pure debt and pure equity. Their classification (debt or equity) can differ for accountants, rating agencies, tax authorities and lenders.

29.3 Dividend policy and returning cash

Policy	Description	Typical company
Residual	Pay out what is left after funding all positive-NPV projects	Theoretically optimal; volatile dividends
Stable / progressive	Steady dividend per share growing gradually	Mature listed companies, utilities
Target payout ratio	Fixed % of earnings	Banks, many family companies
Special dividends	One-off distributions of excess cash	After asset sales
Share buybacks	Repurchase shares	Flexible; tax-efficient in some jurisdictions
No dividend	Reinvest everything	Growth companies, startups

Constraints: company law (distributable reserves, solvency tests), loan covenants (dividend lock-ups when leverage is high), regulatory capital (banks, insurers), cash availability at the holding company, tax on dividends and withholding taxes for foreign shareholders.

بالعربية — سياسة توزيع الأرباح

لا توجد سياسة توزيع واحدة صحيحة لكل الشركات. الشركات الناضجة ذات التدفقات المستقرة توزّع نسبة كبيرة من أرباحها، بينما تحتفظ الشركات النامية بأرباحها لتمويل التوسع. ويجب مراعاة القيود القانونية (الاحتياطيات القابلة للتوزيع) والتعهدات البنكية التي قد تمنع التوزيع عند ارتفاع المديونية، إضافة إلى احتياجات النقد المستقبلية.

PRACTICE · Practice questions

1. Peer equity beta 1.1 at D/E 0.5, tax 20%. Your target D/E is 1.0. Compute the unlevered and relevered beta.
2. Ke 14%, Kd 8% pre-tax, tax 25%, target D/(D+E) = 40%. Compute WACC.
3. A family company earns ROA of 7% pre-tax and can borrow at 9%. The owners want to increase leverage to “boost returns”. What do you tell them?

ANSWER · Answers to the practice questions

1. $\beta U = 1.1 \div [1 + 0.8 \times 0.5] = 1.1 \div 1.4 =$ **0.786**. $\beta L = 0.786 \times [1 + 0.8 \times 1.0] =$ **1.414**.
2. $0.6 \times 14\% + 0.4 \times 8\% \times 0.75 = 8.4\% + 2.4\% =$ **10.8%**.
3. Borrowing at 9% to earn 7% (pre-tax) reduces ROE and increases risk — negative leverage. (After tax, the comparison is ROA × (1 − t) vs Kd × (1 − t); the conclusion is the same.) Improve operating returns first.

CHAPTER 30

Working Capital and Cash Management

LEARNING OBJECTIVES · In this chapter

Managing receivables, inventory and payables; working-capital policies and financing; the 13-week cash flow forecast; and the economics of releasing cash.

30.1 Why working capital deserves senior attention

Working capital is usually the largest controllable source (or use) of cash in trading, distribution and manufacturing companies. Reducing the cash conversion cycle by 10 days on $200m revenue releases about $5.5m — often more than a year's net profit for a distributor.

30.2 Working-capital policies

Policy	Investment in current assets	Financing	Profitability	Risk
Conservative	High (large buffers of stock and cash)	Mostly long-term funds	Lower	Lower
Moderate (matching)	Moderate	Permanent current assets funded long-term; seasonal peaks short-term	Balanced	Balanced
Aggressive	Lean	Heavy short-term funding	Higher	Higher (refinancing and stock-out risk)

The **permanent** part of working capital (the minimum level needed all year) should be financed with long-term or committed funding; **seasonal** peaks with revolving or short-term facilities.

30.3 Receivables management

Levers: credit policy (who gets credit and how much), terms (30/60/90 days), invoicing speed and accuracy, collections process, dispute resolution, incentives (early-payment discounts), and risk transfer (credit insurance, factoring, L/Cs).

WORKED EXAMPLE · Should we offer an early-payment discount?

Customers pay on average in 75 days. Offer 1.5% discount for payment within 15 days; expect 40% of $50m annual sales to take it. - Cost: 1.5% × 20m = **$300k** per year. - Benefit: receivables reduced by 20m × (75 − 15)/365 = **$3.29m**; at a 9% funding cost that saves **$296k**; plus lower bad-debt risk and collection effort. - Roughly break-even financially; worthwhile only if credit risk or liquidity is a concern, or if funding is constrained. (The implied annualised cost of the discount is about 9.3% simple: 1.5/98.5 × 365/60.)

30.4 Inventory management

FORMULA · Economic order quantity (EOQ)

EOQ = √(2 × D × S ÷ H)

D = annual demand (units); S = cost per order; H = holding cost per unit per year (financing, storage, insurance, obsolescence — often 15–30% of unit cost per year).

Example: D = 12,000 units, S = $150, H = $4 → EOQ = √(900,000) ≈ **949 units** (about 13 orders a year). EOQ is a starting point; minimum order quantities, container sizes, price breaks and lead-time variability adjust it in practice.

Other tools: ABC analysis (A items = ~80% of value, managed tightly), safety stock and reorder points (see Part IX), vendor-managed inventory, consignment stock, just-in-time, and periodic review of slow-moving stock.

30.5 Payables management

Extending payables is the cheapest finance — until it damages supply. Good practice: negotiate terms rather than paying late; pay on the due date, not early (unless a discount justifies it); centralise payment runs; use supply chain finance so suppliers can get paid early at the buyer's credit cost.

30.6 Financing working capital

Instrument	How it works	Suited to
Overdraft	On-demand limit on current account	Small timing gaps
Revolving credit facility	Committed draw/repay facility	Mid-size and large companies
Invoice discounting / factoring	Advance of 70–90% of eligible receivables; factoring may include collection and non-recourse risk transfer	Companies with good-quality receivables
Borrowing-base facility	Limit = eligible receivables × advance rate + eligible inventory × advance rate	Traders, distributors
Inventory / warehouse finance	Loans against stock, often with collateral management	Commodities, distributors
Import loans / trust receipts	Bank pays supplier (often under an L/C) and lends for 90–180 days until goods are sold	Importers
Supply chain finance (reverse factoring)	Bank pays the buyer's suppliers early based on approved invoices	Large buyers with many suppliers
Pre-export finance	Loan to an exporter repaid from export proceeds	Commodity producers

WORKED EXAMPLE · Borrowing base (fictional distributor)

Item	Amount	Eligibility / advance rate	Availability
Receivables	$8.0m	Exclude >90 days ($0.6m) and concentrations above 20% ($0.4m); advance 80% on $7.0m	$5.60m
Inventory	$6.0m	Exclude slow-moving ($1.0m); advance 50% on $5.0m	$2.50m
Borrowing base			**$8.10m**
Facility limit			$10.0m

Item	Amount	Eligibility / advance rate	Availability
Available to draw		Lower of base and limit	**$8.10m**

Borrowing-base reports are typically submitted monthly; the bank audits the underlying data periodically.

30.7 Cash management and the 13-week cash flow forecast

The **13-week cash flow forecast (13WCF)** is a weekly, direct-method forecast of receipts and payments. It is the core tool of treasury in tight periods and is required by lenders in any restructuring.

WORKED EXAMPLE · 13-week forecast excerpt (fictional, $000)

Week	1	2	3	4	…	13
Opening cash	850	610	1,020	330	…	
Customer receipts	1,200	1,650	900	1,400	…	
Supplier payments	(1,050)	(800)	(1,100)	(950)	…	
Payroll	—	—	(420)	—	…	
Rent, utilities, other	(90)	(60)	(70)	(60)	…	
Debt service	(300)	—	—	—	…	
Tax / VAT	—	(380)	—	—	…	
Closing cash	**610**	**1,020**	**330**	**720**	…	
Facility headroom	1,500	1,500	1,500	1,500	…	
Liquidity	**2,110**	**2,520**	**1,830**	**2,220**	…	

Practice: update weekly, compare forecast vs actual (variance analysis), and investigate large differences; the accuracy of the forecast becomes a measure of management credibility with lenders.

PRACTICE · Practice questions

1. A company with revenue $120m and COGS $90m has DSO 70, DIO 80, DPO 40. Management targets DSO 55, DIO 65, DPO 50. How much cash is released?
2. Compute EOQ for D = 50,000 units, S = $200, unit cost $25, holding cost 20% per year.
3. Receivables $5m (of which $0.8m over 90 days, and one customer of $1.5m where the concentration limit is 20% of eligible receivables), inventory $3m (slow-moving $0.4m), advance rates 80% and 50%. Compute the borrowing base.

ANSWER · Answers to the practice questions

1. DSO −15 days × 120/365 = $4.93m; DIO −15 × 90/365 = $3.70m; DPO +10 × 90/365 = $2.47m → total ≈ **$11.1m** released.
2. H = 0.2 × 25 = $5 → EOQ = √(2 × 50,000 × 200 ÷ 5) = √4,000,000 = **2,000 units**.
3. Eligible receivables before concentration: 5.0 − 0.8 = 4.2; 20% limit = 0.84 → exclude 1.5 − 0.84 = 0.66 → eligible 3.54 → × 80% = **2.83**. Inventory: (3.0 − 0.4) × 50% = **1.30**. Base ≈ **$4.13m** (iteration: concentration limits are sometimes applied to total eligible after the excess is removed; conventions vary by facility).

CHAPTER 31

Financial Planning, Forecasting and Modelling

LEARNING OBJECTIVES · In this chapter

Forecasting methods, the external funding requirement of growth, the architecture of a three-statement financial model, modelling best practice and how analysts use code alongside spreadsheets.

31.1 Planning horizons

Horizon	Tool	Owner	Purpose
3–5 years	Strategic plan / long-range model	CEO, CFO, board	Strategy, capital allocation, financing
12 months	Annual budget	All budget owners	Targets, resource allocation
Rolling 12–18 months	Rolling forecast	FP&A	Updated expectation
13 weeks	Cash flow forecast	Treasury	Liquidity
Transaction-specific	Deal model / project model	Corporate development, bankers	Investment, lending, valuation

31.2 Forecasting methods

- **Top-down:** market size × market share.
- **Bottom-up / driver-based:** customers × orders × average order value; capacity × utilisation × price; headcount × productivity. Driver models expose assumptions and are easier to challenge.
- **Percentage-of-sales:** many costs and balance-sheet items scale with revenue — quick but crude.
- **Time-series and statistical models:** moving averages, exponential smoothing, regression, seasonality decomposition; machine-learning models for demand at SKU level (see Part XVI).
- **Judgemental / pipeline-based:** weighted sales pipeline for B2B.

PROFESSIONAL INSIGHT · The professional's rule for forecasts

A forecast is a set of explicit, defensible assumptions — not a number. Show the drivers, compare them with history and with peers, and identify the two or three assumptions that matter most. "Revenue grows 25%" is weak; "we add 40 hotel customers at an average of $60k per year, consistent with last year's 35, with churn of 8% as in the last three years" is strong.

31.3 How much external funding does growth need?

FORMULA · Additional funds needed (AFN), percentage-of-sales method

$$\text{AFN} = (A^* \div S_0) \times \Delta S - (L^* \div S_0) \times \Delta S - M \times S_1 \times (1 - \text{payout})$$

A^* = assets that grow with sales; L^* = liabilities that grow spontaneously with sales (payables, accruals); M = net margin; S_1 = next year's sales.

WORKED EXAMPLE · AFN for a fictional distributor

S_0 = \$50m; growth 20% → ΔS = \$10m; S_1 = \$60m. A/S_0 = *60%; L*/S_0 = 15%; net margin 5%; payout 40%. AFN = 0.60 × 10 − 0.15 × 10 − 0.05 × 60 × 0.60 = 6.0 − 1.5 − 1.8 = **\$2.7m**. At 30% growth: ΔS = 15 → AFN = 9.0 − 2.25 − 0.05 × 65 × 0.6 = **\$4.8m**. The faster the growth, the more external funding is required — banks must be engaged before growth, not after the overdraft is full.

31.4 The three-statement model

A three-statement model links forecast income statement, balance sheet and cash flow statement so that every assumption flows consistently and the balance sheet balances.

Module	Contents
Inputs / assumptions	Volumes, prices, cost ratios, working-capital days, capex plan, depreciation lives, debt terms, tax, dividends, scenarios
Revenue build	Drivers by segment/product/region
Operating costs	Fixed and variable costs, headcount
Working capital	DSO, DIO, DPO → balances
Fixed assets	Opening + capex − depreciation − disposals
Debt schedule	Opening, drawdowns, repayments, interest (often on average balance → circularity), revolver as plug
Tax	Taxable profit, tax losses, cash tax timing
Equity	Opening + net income − dividends + issues
Statements	Income statement, balance sheet, cash flow (indirect)
Outputs	KPIs, ratios, covenants, valuation, charts, scenario comparison
Checks	Balance check = 0; cash flow ending cash = balance sheet cash; covenant tests

The usual **balancing mechanism**: forecast all items except cash and the revolving facility; the cash flow statement produces ending cash; if cash would fall below a minimum, the revolver draws (a “plug”); if surplus, it repays.

Modelling best practice

- Separate inputs (clearly coloured, one location), calculations and outputs.
- One formula per row, consistent across columns; no hard-coded numbers inside formulas.
- Timeline flags (e.g., operating period, construction period) instead of manually edited ranges.
- Sign conventions documented; units in every row label.
- Error checks visible on every sheet; circularity controlled with a switch.
- Version control and change logs; independent model review for large decisions.
- Keep it as simple as the decision allows.

COMMON MISTAKES · Common modelling errors

- Balance sheet does not balance and someone adds a “plug” to equity.
- Working capital forecast as a fixed amount while revenue doubles.
- Capex below depreciation for years while volumes grow (assets would wear out).
- Interest calculated on the opening debt only when large drawdowns occur in the year.
- Mixing nominal growth with real costs (inflation inconsistency).
- Terminal year not at a steady state (margins still expanding, capex below depreciation).

31.5 Code alongside spreadsheets

Spreadsheets dominate business communication; code is better for large data, repeated runs and simulation. A finance–CS professional uses both: Python or SQL to prepare data and run simulations, a clean spreadsheet or BI report to communicate.

```
# Monte Carlo simulation of the cold-storage project NPV (illustrative)
import numpy as np
rng = np.random.default_rng(42)
n = 100_000
ebitda = rng.normal(650, 90, n)            # $000 per year
capex  = rng.triangular(1900, 2000, 2300, n)
rate, tax, years = 0.11, 0.20, 5
dep = capex / years
annual = (ebitda - dep) * (1 - tax) + dep  # after-tax cash flow
annuity = (1 - (1 + rate) ** -years) / rate
terminal = (240 + 200) / (1 + rate) ** years  # after-tax salvage + WC recovery
npv = -(capex + 200) + annual * annuity + terminal
print(f"mean NPV {npv.mean():.0f}k, P(NPV<0) {np.mean(npv < 0):.1%}")
```

The output is a distribution — e.g., a mean NPV near the base case but with a meaningful probability of a negative NPV — which supports a richer discussion than a single number.

PRACTICE · Practice questions

1. Using the AFN formula: S_0 \$80m, growth 15%, A/S_0 55%, L/S_0 20%, net margin 4%, payout 50%. Compute AFN.
2. A model shows revenue growing from \$20m to \$60m in five years with working capital constant at \$4m. What is wrong and how would you fix it?
3. Why do many models use the revolving facility as the balancing item?

ANSWER · Answers to the practice questions

1. $\Delta S = 12$; $S_1 = 92$. AFN = $0.55 \times 12 - 0.20 \times 12 - 0.04 \times 92 \times 0.5 = 6.6 - 2.4 - 1.84 =$ **\$2.36m**.
2. Working capital should scale with activity: model receivables, inventory and payables from days (DSO/DIO/DPO) on forecast revenue and COGS. At constant days, WC would roughly triple to ~\$12m, requiring ~\$8m of extra funding the model ignores.
3. It mirrors reality: companies fund shortfalls by drawing on committed lines and repay with surpluses; it lets the model close (cash and debt both determined) and shows peak funding needs and covenant pressure.

CHAPTER 32

Leverage, Break-even, Sensitivity and Scenario Analysis

LEARNING OBJECTIVES · In this chapter

Break-even analysis, operating and financial leverage, sensitivity and scenario analysis, simulation, and the main categories of financial risk.

32.1 Break-even analysis

FORMULA · Break-even

Break-even units = Fixed costs ÷ (Price − Variable cost per unit)
Break-even revenue = Fixed costs ÷ Contribution margin ratio
Target-profit units = (Fixed costs + Target profit) ÷ Contribution per unit
Margin of safety = (Actual sales − Break-even sales) ÷ Actual sales

WORKED EXAMPLE · Fictional furniture manufacturer

Fixed costs \$1.2m per year; price \$50; variable cost \$30 → contribution \$20 (CM ratio 40%). - Break-even = 1.2m ÷ 20 = **60,000 units** = **\$3.0m revenue**. - Current sales 80,000 units → margin of safety = 20,000 ÷ 80,000 = **25%**. - To earn \$0.6m profit: (1.2 + 0.6) ÷ 20 = **90,000 units**.

32.2 Operating, financial and total leverage

FORMULA · Degrees of leverage

DOL = Contribution ÷ EBIT
DFL = EBIT ÷ (EBIT − Interest)
DTL = DOL × DFL = Contribution ÷ (EBIT − Interest)

Continuing the example at 80,000 units: contribution \$1.6m; EBIT \$0.4m; interest \$0.1m. DOL = 1.6 ÷ 0.4 = **4.0**; DFL = 0.4 ÷ 0.3 = **1.33**; DTL = **5.33**. A 10% fall in volume reduces EBIT by 40% (to \$0.24m) and profit before tax by 53% (to \$0.14m).

Business	Operating leverage	Financial leverage	Combined risk
Asset-light broker	Low	Low	Low
Airline	High	Often high	Very high
Utility	High	High, but stable regulated revenue	Moderate
Software (mature)	High	Low to moderate	Moderate
Leveraged buyout of a distributor	Low-moderate	High	High

32.3 Sensitivity analysis

Change one input at a time and observe the output. Present results as a table or **tornado chart** (inputs ranked by impact).

WORKED EXAMPLE · Sensitivity of the cold-storage project NPV (base NPV $279k)

Input changed	Low case	NPV ($000)	High case	NPV ($000)
Annual EBITDA ±20%	520	(106)	780	663
Capex ±10%	2,200	108	1,800	449
Discount rate ±1 pt	12%	213	10%	348
Salvage value ±$150k	150	207	450	350

EBITDA is the dominant driver; the investment decision should focus on the reliability of third-party storage demand and pricing, not on fine-tuning the discount rate.

32.4 Scenario analysis

Scenarios change several related inputs together to describe coherent states of the world.

Scenario	Narrative	Probability	NPV ($000)
Upside	Anchor tenant signs; tariffs rise with food inflation	25%	700
Base	Plan as modelled	50%	300
Downside	Recession; occupancy 60%; energy costs +30%	25%	(400)
Probability-weighted			**225**

Use scenarios to ask: Can the company survive the downside (covenants, liquidity)? What early indicators tell us which scenario is unfolding? What actions (pre-letting, fixed energy contracts, phased construction) improve the downside?

32.5 Financial risk categories

Risk	Description	Typical measures and tools
Market risk	Losses from FX, interest rates, commodity and equity price moves	Hedging, matching, limits, VaR
Credit risk	Counterparty fails to pay	Credit limits, insurance, collateral, L/Cs
Liquidity risk	Cannot meet obligations or refinance	Cash buffers, committed facilities, maturity spread
Funding / refinancing risk	Debt matures when markets are closed or rates are high	Early refinancing, diversified sources
Operational risk	Process, people, system failures	Controls, insurance, BCP
Covenant risk	Breach triggers default	Headroom monitoring, early negotiation

PRACTICE · Practice questions

1. Fixed costs $900k, price $120, variable cost $75. Compute break-even units and revenue, and units needed for $270k profit.
2. EBIT $2m, interest $0.8m, contribution $6m. Compute DOL, DFL and DTL. What happens to PBT if sales fall 5%?
3. Why might a project with a positive expected NPV still be rejected?

ANSWER · Answers to the practice questions

1. Contribution \$45 → BE = 900,000 ÷ 45 = **20,000 units**; revenue **\$2.4m**; for \$270k profit: 1,170,000 ÷ 45 = **26,000 units**.
2. DOL = 3; DFL = 2 ÷ 1.2 = 1.67; DTL = **5**. Sales −5% → PBT ≈ −25% (from \$1.2m to \$0.9m).
3. The downside may threaten the company's survival (liquidity, covenants), the risk may be outside appetite, the capital may be better used elsewhere (higher NPV per dollar), or strategic and execution risks are not captured in the model.

CHAPTER 33

Business Valuation

LEARNING OBJECTIVES · In this chapter

Enterprise value vs equity value; trading comparables; precedent transactions; discounted cash flow valuation with a complete example; valuation of private companies; and how professionals triangulate.

33.1 What is being valued?

FORMULA · Enterprise value to equity value bridge

Equity value = Enterprise value − Debt − Debt-like items (leases if EBITDA is post-IFRS 16, pension deficits, certain provisions, deferred consideration) − Non-controlling interests − Preferred equity + Cash + Non-operating assets (investments, associates, surplus property)
Equity value per share = Equity value ÷ diluted shares outstanding

Consistency rule: enterprise-value multiples (EV/EBITDA, EV/EBIT, EV/Sales) use metrics available to all capital providers (before interest); equity-value multiples (P/E, P/B) use metrics after interest (net income, book equity). Mixing them is a classic error.

33.2 Approach 1: Trading comparables ("comps")

1. Select peers: same industry, business model, size, geography, growth.
2. Collect market values (market cap, net debt) and financial metrics (LTM and forecast).
3. Compute multiples; examine the range; adjust for differences.
4. Apply the relevant multiple to the target's metric.

WORKED EXAMPLE · Comps for a fictional regional logistics company (EBITDA $20m)

Peer	EV ($m)	EBITDA ($m)	EV/EBITDA	Revenue growth	EBITDA margin
Peer A	450	60	7.5×	6%	14%
Peer B	820	100	8.2×	9%	16%
Peer C	540	60	9.0×	12%	18%
Peer D	272	40	6.8×	4%	11%
Peer E	1,010	100	10.1×	15%	20%
Median			**8.2×**		

Interquartile range 7.5–9.0× → EV **$150–180m**, median $164m. If the target grows faster than the median peer with higher margins, the upper half of the range is defensible; for a smaller, less liquid company, a discount may apply.

33.3 Approach 2: Precedent transactions

Multiples paid in past acquisitions of comparable companies. They include a **control premium** (buyers pay for control and synergies) and reflect market conditions at the time. Weaknesses: limited and dated data, different deal structures, undisclosed terms.

Example: median precedent EV/EBITDA 9.5× → EV ≈ **$190m** for the target above.

33.4 Approach 3: Discounted cash flow (DCF)

FORMULA · DCF (enterprise value)

$EV = \Sigma\ FCFF_t \div (1 + WACC)^t + TV_n \div (1 + WACC)^n$

Gordon growth terminal value: $TV_n = FCFF_n \times (1 + g) \div (WACC - g)$

Exit multiple terminal value: $TV_n = EBITDA_n \times$ exit multiple

WORKED EXAMPLE · DCF of the same fictional company

Forecast free cash flow to the firm ($m): Year 1: 10.0; Year 2: 11.0; Year 3: 12.0; Year 4: 12.8; Year 5: 13.5. WACC 10%; terminal growth 3%; net debt $40m; 10m shares.

Year	1	2	3	4	5
FCFF	10.0	11.0	12.0	12.8	13.5
Discount factor (10%)	0.9091	0.8264	0.7513	0.6830	0.6209
PV	9.09	9.09	9.02	8.74	8.38

- Sum of PV of forecast FCFF = **$44.3m**.
- $TV_5 = 13.5 \times 1.03 \div (0.10 - 0.03)$ = **$198.6m**; PV = 198.6 × 0.6209 = **$123.3m**.
- **EV = $167.7m**; equity = 167.7 − 40 = **$127.7m**; **$12.77 per share**.
- The terminal value is 74% of EV — typical, and a reason to scrutinise long-term assumptions.
- Cross-check with an exit multiple of 8.0× year-5 EBITDA of $20m: TV = $160m; PV $99.3m; EV $143.7m. The Gordon-growth TV implies an exit multiple of about 9.9× — on the high side relative to peers.

Sensitivity of EV ($m) to WACC and terminal growth:

WACC \ g	2%	3%	4%
9%	173.4	196.2	228.0
10%	151.2	167.7	189.6
11%	134.0	146.3	162.2

Small changes in WACC and g move value by ±20–35%. A DCF is a structured way to think about value drivers, not a precise answer.

33.5 Triangulation: the “football field”

Method	Low EV ($m)	High EV ($m)
Trading comps (7.5–9.0×)	150	180
Precedent transactions (8.5–10.5×)	170	210
DCF (WACC 9–11%, g 2–4%)	134	228
DCF central case	146	190
Indicative range	**155**	**185**

The final judgement weighs which methods are most reliable for this company and purpose (a minority investment vs a control acquisition).

33.6 Valuing private companies

Private companies are typically valued with the same methods plus adjustments:

- **Normalisation of earnings:** owner salaries above or below market, related-party rents, one-off items, personal expenses run through the company.
- **Discount for lack of marketability (DLOM)** for minority stakes that cannot be easily sold.
- **Size and specific-risk premiums** in the discount rate.
- **Quality of earnings** review: are profits real, recurring and converting to cash?
- **Working-capital normalisation:** in acquisitions, a target level of working capital ("peg") is agreed; deviations at closing adjust the price.

COMMON MISTAKES · Common valuation mistakes

- Applying EV multiples to net income or P/E to EBITDA.
- Using peers with very different growth, margins or risk without adjustment.
- Forgetting debt-like items (leases, deferred consideration, unpaid taxes, end-of-service obligations).
- DCF terminal growth above long-term nominal GDP growth, or terminal year capex below depreciation.
- WACC in one currency, cash flows in another.
- Treating a valuation as a price: price is what a buyer pays after negotiation.

PRACTICE · Practice questions

1. Market cap \$600m; debt \$250m; cash \$50m; lease liabilities \$40m; NCI \$30m; investment in associate \$20m. Compute EV.
2. A company has net income \$12m and EBITDA \$30m; peers trade at 14× P/E and 8× EV/EBITDA; net debt \$70m. Compute implied equity values from both methods and comment.
3. FCFF in year 5 is \$25m; WACC 9%; g 2.5%. Compute TV and its PV (discount factor 0.6499).

ANSWER · Answers to the practice questions

1. EV = 600 + 250 + 40 + 30 − 50 − 20 = **\$850m** (leases included if EBITDA is post-IFRS 16).
2. P/E: 14 × 12 = **\$168m**. EV/EBITDA: 8 × 30 = 240 − 70 = **\$170m**. Consistent; if they diverged, check for unusual interest, tax or leverage differences versus peers.
3. TV = 25 × 1.025 ÷ 0.065 = **\$394.2m**; PV = 394.2 × 0.6499 = **\$256.2m**.

CHAPTER 34

Mergers, Acquisitions and Corporate Restructuring: The Financial View

LEARNING OBJECTIVES · In this chapter

How buyers decide what to pay, how synergies are valued, how deals are financed and structured, accretion/dilution, leveraged buyouts, and the finance of restructuring. Legal aspects are covered in Part XII and integration in Part XIV.

34.1 Value creation in an acquisition

FORMULA · Value created for the acquirer

Value to acquirer = Standalone value of target + PV of synergies − Price paid − Transaction and integration costs

The control premium paid = Price − standalone market value; the acquirer gains only if synergies exceed the premium plus costs.

Synergy type	Examples	Reliability
Cost synergies	Procurement savings, closing duplicate offices, shared IT	Higher — within management's control
Revenue synergies	Cross-selling, new markets, pricing	Lower — depend on customers
Financial synergies	Lower funding cost, tax efficiencies	Varies; scrutinise
Dis-synergies	Customer losses, staff attrition, integration costs	Often underestimated

Buyers typically value synergies net of integration costs, phased in over 2–3 years, discounted, and share only part of them with the seller in the price.

34.2 Deal structures and consideration

Element	Options	Considerations
What is bought	Shares (company with all its history) or assets (selected assets/liabilities)	Liabilities inherited; tax; licences and contracts transferability
Consideration	Cash, shares, mix, deferred consideration, earn-outs, vendor loans	Risk sharing; seller's tax; acquirer's balance sheet
Price mechanism	Completion accounts (price adjusted for actual net debt and working capital at closing) or locked box (price fixed on a historical balance sheet with leakage protection)	Certainty vs accuracy
Protection	Warranties, indemnities, escrow, W&I insurance	Allocation of unknown risks

Earn-outs link part of the price to future performance (e.g., EBITDA in years 1–2), bridging valuation gaps but often causing disputes over accounting and management control.

34.3 Accretion and dilution

Listed acquirers analyse the effect on earnings per share (EPS).

WORKED EXAMPLE · Accretion/dilution (fictional)

Acquirer: net income $100m, 50m shares, EPS $2.00, share price $30 (P/E 15). Target: net income $20m; price $300m (P/E 15).

- **All-debt financing** at 6% pre-tax, tax 20%: after-tax interest = 300 × 6% × 0.8 = $14.4m. New EPS = (100 + 20 − 14.4) ÷ 50 = **$2.112 (+5.6% accretive)**.
- **All-stock financing** at $30: new shares 10m. New EPS = 120 ÷ 60 = **$2.00 (neutral)**.
- Add after-tax synergies of $4m: debt case EPS $2.192; stock case $2.067.

Rule of thumb: buying a company on a lower P/E than your own with stock is accretive; with debt, the deal is accretive if the target's earnings yield (1/P/E paid) exceeds the after-tax cost of debt. **Accretion does not equal value creation** — paying too much financed with cheap debt can be EPS-accretive and still destroy value.

34.4 Leveraged buyouts (LBOs)

A financial sponsor (private equity) acquires a company using significant debt, repaid from the target's cash flows, and sells it after 3–7 years.

WORKED EXAMPLE · LBO returns (fictional)

- Entry: EBITDA $50m at 8.0× → EV $400m; debt 5.0× = $250m; sponsor equity $150m (fees ignored).
- Over 5 years: EBITDA grows to $65m; free cash flow repays debt to $100m.
- Exit at 8.0× → EV $520m; equity = 520 − 100 = **$420m**.
- MOIC = 420 ÷ 150 = **2.8×**; IRR = 2.8^(1/5) − 1 ≈ **22.9%**.

Value drivers: EBITDA growth (operational improvement), debt paydown (deleveraging), and multiple expansion (buying cheaper than selling — not something to rely on). High leverage also creates the risk of covenant breach if EBITDA falls.

34.5 Financing an acquisition

Cash on hand; acquisition term loans and bridge loans (later refinanced with bonds or equity); vendor financing; equity issue; seller rollover (seller keeps a minority stake). Lenders focus on pro forma leverage, interest cover, the credibility of synergies, and integration risk.

34.6 Corporate restructuring: the financial mechanics

When a company cannot sustain its debt, stakeholders negotiate a restructuring:

1. **Stabilise liquidity:** standstill, 13-week cash flow, cost and capex cuts, working-capital release.
2. **Independent business review:** viability, sustainable debt capacity (e.g., the debt the business can service at 3–4× normalised EBITDA).
3. **Capital structure options:** maturity extensions, covenant resets, interest holidays or payment-in-kind, partial write-offs, debt-for-equity swaps, new money (often super-senior), asset disposals, new equity from shareholders.
4. **Allocation of losses:** by priority — equity first, then subordinated, then senior. The **absolute priority** principle is a guide; negotiated deals often give junior parties something to secure agreement.
5. **Implementation:** consensual agreement or court-supported procedure (jurisdiction-specific).

WORKED EXAMPLE · Debt-for-equity swap (fictional)

Company with sustainable EBITDA $30m carries $300m of debt (10×). Sustainable debt estimated at 4× = $120m. Lenders exchange $180m of debt for 80% of the equity; existing shareholders keep 20% and inject $20m of new money. Lenders lose part of their claim but gain the upside of a viable company; shareholders avoid total loss.

PRACTICE · Practice questions

1. Target standalone value $500m; synergies PV $120m; integration costs $30m; price $580m. Does the deal create value for the acquirer? How much of the synergy value goes to the seller?
2. An acquirer on P/E 12 buys a target on P/E 18 using all stock. Accretive or dilutive (ignore synergies)?
3. LBO: entry EV 10× EBITDA of $40m with 6× debt. Five years later EBITDA $48m, debt $180m, exit multiple 9×. Compute MOIC and approximate IRR.

ANSWER · Answers to the practice questions

1. Value to acquirer = 500 + 120 − 580 − 30 = **+$10m** — marginally positive. Premium paid = 80; the seller captures 80 of the 120 synergies (67%), leaving little margin for error.
2. **Dilutive**: the acquirer issues shares valued at 12× its earnings to buy earnings at 18×; earnings yield acquired (5.6%) < acquirer's earnings yield (8.3%).
3. Entry EV 400; debt 240; equity 160. Exit EV 9 × 48 = 432; equity 432 − 180 = 252. MOIC = 252 ÷ 160 = **1.58×**; IRR ≈ 1.575^(0.2) − 1 ≈ **9.5%** — weak: multiple contraction and modest growth.

SUMMARY · Part IV summary

- Value comes from cash flows, their timing and their risk; NPV is the core rule, IRR and payback are supporting tools.
- The cost of capital is an opportunity cost set by markets, estimated with judgement and presented as a range.
- Capital structure balances cheaper debt against distress risk and flexibility; leverage magnifies outcomes in both directions.
- Working capital is a major, controllable source of cash; growth needs funding before it happens.
- Valuation triangulates DCF, comparables and precedents; the assumptions matter more than the arithmetic.
- In M&A, value is created only if synergies exceed the premium and costs — EPS accretion is not proof of value.

PART V

Economics and the Macroeconomic Business Environment

الاقتصاد وبيئة الأعمال الكلية

Every company operates inside an economy it cannot control. Prices of inputs, demand from customers, interest rates on borrowings, the value of the currency, access to foreign exchange, tariffs, sanctions and political stability all change the numbers in the business plan. This part explains the economic machinery and — most importantly — translates it into company-level consequences with numerical examples.

LEARNING OBJECTIVES · Learning objectives for Part V

- Apply supply, demand and elasticity to pricing and market analysis.
- Understand market structures and their implications for competition and margins.
- Interpret GDP, inflation, employment and business-cycle indicators.
- Explain how central banks set interest rates and how policy decisions reach companies.
- Analyse exchange rates, the balance of payments, tariffs and trade barriers.
- Assess geopolitical, sanctions, country and sovereign risks.
- Quantify the effect of inflation, interest-rate and currency changes on importers, exporters, borrowers and investors.

Chapters in this part

Chapter 35 Microeconomics for Business: Markets, Prices and Competition 32
Chapter 36 Macroeconomics: Growth, Inflation, Employment and the Business Cycle 35
Chapter 37 Money, Interest Rates, Central Banks and Government Finance 37
Chapter 38 Exchange Rates, the Balance of Payments and International Trade Policy 40
Chapter 39 Geopolitical, Sanctions, Country and Sovereign Risk — and How Macro Shocks Reach Companies 43

CHAPTER 35

Microeconomics for Business: Markets, Prices and Competition

LEARNING OBJECTIVES · In this chapter

Supply and demand, equilibrium, elasticity, production and productivity, costs, and market structures — applied to business decisions.

35.1 Supply, demand and equilibrium

Demand is the quantity buyers are willing and able to purchase at each price; it usually falls as price rises. **Supply** is the quantity producers are willing to sell at each price; it usually rises with price. The **equilibrium** price clears the market.

Shift	Cause examples	Effect on price and quantity
Demand increases	Higher incomes, population growth, new uses, lower price of complements	Price ↑, quantity ↑
Demand decreases	Recession, substitutes cheaper, changing tastes	Price ↓, quantity ↓
Supply increases	New capacity, better technology, lower input costs	Price ↓, quantity ↑
Supply decreases	Drought, war, plant outages, export bans, sanctions	Price ↑, quantity ↓

WORKED EXAMPLE · Reading a commodity market (fictional numbers)

A drought cuts wheat output in a major exporting region by 8%. Short-run demand for wheat is inelastic (people still eat bread). Price elasticity of demand −0.2 implies price must rise about 8% ÷ 0.2 = **40%** to reduce consumption by 8% if no stocks are released. In practice, stocks, substitution (other grains for feed) and government stock releases moderate the rise — but the example explains why commodity prices spike dramatically on modest supply shocks.

35.2 Elasticity

FORMULA · Elasticities

Price elasticity of demand: $Ed = \%\Delta Q \div \%\Delta P$ (usually negative; $|Ed| > 1$ elastic, < 1 inelastic)
Income elasticity: $\%\Delta Q \div \%\Delta \text{Income}$ (> 1 luxury; 0–1 necessity; < 0 inferior good)
Cross-price elasticity: $\%\Delta Q$ of A ÷ $\%\Delta P$ of B (> 0 substitutes; < 0 complements)
Revenue effect of a price change $\approx \%\Delta P \times (1 + Ed)$ for small changes

Business implications:

- **Pricing:** if demand for your product is inelastic (few substitutes, small share of customer's budget, high switching costs), price increases raise revenue.
- **Tax and tariff incidence:** when demand is inelastic, the buyer bears most of a tax or tariff; when elastic, the seller absorbs more.
- **Forecasting:** income elasticity tells you which products suffer most in recessions (luxury goods, construction materials) and which are defensive (basic food, utilities).

35.3 Production, productivity and costs

Productivity = output ÷ input (labour productivity: output per hour; total factor productivity: output growth not explained by inputs). Productivity growth is the main source of long-run wage and profit growth.

Short run: some inputs are fixed; **diminishing marginal returns** set in. Long run: all inputs variable; **economies of scale** (falling average cost) up to a point, then possible diseconomies (coordination costs).

Cost concept	Definition	Decision use
Marginal cost	Cost of one more unit	Pricing floor for incremental orders
Average total cost	Total cost ÷ units	Long-run pricing, capacity
Average variable cost	Variable cost ÷ units	Short-run shutdown decision: produce if price > AVC
Sunk cost	Unrecoverable past spending	Ignore
Opportunity cost	Best alternative forgone	Always include

35.4 Market structures

Structure	Number of sellers	Product	Barriers	Pricing power	Examples
Perfect competition	Many	Identical	None	None (price taker)	Some agricultural markets
Monopolistic competition	Many	Differentiated	Low	Some	Restaurants, retail brands
Oligopoly	Few	Identical or differentiated	High	Significant, interdependent	Telecoms, cement, container shipping, aircraft manufacturing
Monopoly	One	Unique	Very high	High (often regulated)	Utilities, some infrastructure
Monopsony / oligopsony	One / few buyers	—	—	Buyer power	Government defence procurement, large retailers vs small suppliers

Concentration is measured by market shares and the **Herfindahl-Hirschman Index** (HHI = sum of squared market shares in %). Competition authorities use HHI levels and changes to screen mergers (thresholds differ by jurisdiction).

WORKED EXAMPLE · HHI

Four firms with shares 40%, 30%, 20%, 10%: HHI = 1,600 + 900 + 400 + 100 = **3,000** — highly concentrated by common thresholds. A merger of the 20% and 10% firms raises HHI by 2 × 20 × 10 = **400** points, which would attract scrutiny in many jurisdictions.

بالعربية — هياكل السوق وأثرها في الربحية

كلما قلّ عدد المنافسين وارتفعت عوائق الدخول زادت قدرة الشركات على التسعير وتحقيق هوامش أعلى. في المنافسة الكاملة تكون الشركة «آخذة للسعر» ولا تملك قوة تسعيرية، أما في احتكار القلة فتراقب الشركات بعضها وتتأثر قراراتها بقرارات المنافسين. لذلك تهتم سلطات المنافسة بعمليات الاندماج التي ترفع درجة التركّز في السوق.

PRACTICE · Practice questions

1. A 6% price increase reduces volume by 9%. Compute the elasticity and approximate change in revenue.

2. When a government imposes a 10% tariff on imported cement and domestic demand is inelastic, who bears most of the tariff? What happens to local producers?
3. Market shares: 35%, 25%, 15%, 15%, 10%. Compute HHI.

ANSWER · Answers to the practice questions

1. Ed = −9 ÷ 6 = **−1.5** (elastic). Revenue ≈ 1.06 × 0.91 − 1 = **−3.5%**.
2. Mostly domestic buyers (higher prices). Local producers gain pricing power and may raise prices toward the tariff-inclusive import price, increasing their margins.
3. 1,225 + 625 + 225 + 225 + 100 = **2,400**.

CHAPTER 36

Macroeconomics: Growth, Inflation, Employment and the Business Cycle

LEARNING OBJECTIVES · In this chapter

The main macroeconomic indicators, what drives them, how they are published, and how they affect company demand, costs and risks.

36.1 Gross domestic product

FORMULA · GDP by expenditure

$GDP = C + I + G + (X - M)$

C = household consumption; I = investment (business capex, housing, inventories); G = government consumption and investment; X − M = net exports.

GDP can also be measured by production (value added by sector) and income (wages + profits + taxes less subsidies). **Nominal GDP** uses current prices; **real GDP** removes inflation. Growth is reported quarter-on-quarter (sometimes annualised, as in the US) or year-on-year — always check which.

For business, the **composition** matters more than the headline: oil-exporting economies distinguish **oil GDP** and **non-oil GDP**; a retailer cares about household consumption and wages; a cement producer about construction and government investment.

36.2 Inflation

Inflation is the sustained rise in the general price level, measured by indices such as the CPI (consumer prices) and PPI (producer prices). **Core inflation** excludes volatile food and energy. Causes: demand exceeding supply capacity (demand-pull), rising input costs (cost-push), currency depreciation (imported inflation), expectations and wage-price dynamics, and excessive money creation (in extreme cases, hyperinflation).

WORKED EXAMPLE · Inflation and a fixed-price service contract

A facilities-management company signs a 3-year contract at a fixed $1.0m per year. Costs in year 1 are $0.85m (margin 15%). Costs are 70% labour (inflation 6%/yr) and 30% materials (inflation 9%/yr).

Year	Labour	Materials	Total cost	Revenue	Margin
1	595.0	255.0	850.0	1,000	15.0%
2	630.7	278.0	908.7	1,000	9.1%
3	668.5	303.0	971.5	1,000	2.8%

An indexation clause (e.g., annual adjustment by a published CPI index, or a formula 70% wage index + 30% materials index) would have protected the margin. Inflation transfers value from those with fixed-price obligations to receive money to those with fixed-price obligations to pay.

Inflation also: raises nominal interest rates; erodes the real value of cash and receivables; inflates nominal profits while understating the cost of replacing assets (historical-cost depreciation is too low); and complicates comparisons over time.

36.3 Employment and wages

The **unemployment rate** = unemployed (seeking work) ÷ labour force. Low unemployment raises wage pressure and staffing difficulty; high unemployment depresses consumption. Many Gulf economies depend heavily on expatriate labour and run localisation (nationalisation) policies — such as quota systems for national employees — that affect hiring costs and planning; requirements must be checked locally.

36.4 Business cycles and recessions

Economies alternate between **expansion**, **peak**, **contraction** and **trough**. Recessions are often triggered by tightening monetary policy, financial crises, commodity shocks, or demand shocks.

Indicator type	Examples	Use
Leading	PMI new orders, building permits, yield-curve slope, credit growth, stock prices, consumer confidence	Anticipate turning points
Coincident	Industrial production, employment, retail sales, GDP	Confirm current state
Lagging	Unemployment duration, inflation, loan defaults	Confirm after the fact

PROFESSIONAL INSIGHT · How companies prepare for cycles

Experienced CFOs keep liquidity headroom through the cycle, avoid funding long-term assets with short-term debt, prefer variable costs where possible, stress-test budgets for a 15–30% revenue fall, and track leading indicators relevant to their customers (e.g., a building-materials distributor monitors permits, cement sales and government project awards).

36.5 Economic growth in the long run

Long-run growth comes from more labour (population, participation), more capital (investment) and higher productivity (technology, skills, institutions). For investors and companies choosing markets, growth potential, demographics, institutional quality, rule of law, and openness to trade matter more than one year's GDP figure.

PRACTICE · Practice questions

1. Nominal GDP grew 9% and the GDP deflator rose 6%. What was real growth?
2. A distributor's sales track non-oil GDP with an elasticity of 1.5. If non-oil GDP growth slows from 4% to 2%, what happens to expected volume growth?
3. In the fixed-price contract example, design a simple escalation clause and show year-3 revenue with it.

ANSWER · Answers to the practice questions

1. $1.09 \div 1.06 - 1 \approx$ **2.8%**.
2. From about 6% to about **3%** volume growth.
3. Clause: "Contract price adjusts annually by 70% of the change in the national wage index plus 30% of the change in the materials price index." Year-3 adjustment factor $\approx 0.7 \times 1.06^2 + 0.3 \times 1.09^2 = 0.7 \times 1.1236 + 0.3 \times 1.1881 = 0.7865 + 0.3564 =$ **1.1429** → revenue ≈ **$1.143m**; margin ≈ (1,143 − 971.5) ÷ 1,143 ≈ **15%**.

CHAPTER 37

Money, Interest Rates, Central Banks and Government Finance

LEARNING OBJECTIVES · In this chapter

How central banks influence interest rates and credit; how interest-rate changes reach companies and asset values; fiscal policy and government debt; and what all of this means for business decisions.

37.1 Central banks and their mandates

Central banks issue currency, act as bankers to banks and the government, supervise banks (in many countries), operate payment systems, manage reserves, and set monetary policy. Mandates vary: price stability (e.g., the European Central Bank), a dual mandate of maximum employment and stable prices (the US Federal Reserve), or exchange-rate stability (countries with currency pegs).

37.2 How monetary policy works

Tool	Mechanism
Policy rate	Rate at which the central bank lends to or remunerates banks; anchors short-term market rates
Open-market operations	Buying/selling securities or repos to manage liquidity
Reserve requirements	Share of deposits banks must hold at the central bank
Quantitative easing / tightening	Large-scale asset purchases (or run-off) affecting longer-term yields
Forward guidance	Communication about the expected path of policy
Macroprudential tools	Loan-to-value limits, capital buffers, lending restrictions

Transmission: policy rate → interbank and reference rates (SOFR, EIBOR, SAIBOR) → bank lending and deposit rates → borrowing, investment and consumption; also → exchange rate → import prices; → asset prices (bonds, equities, property) → wealth and collateral values; → expectations.

STANDARD / FRAMEWORK · Pegged currencies and monetary policy

Countries that peg their currency to the US dollar — for example Saudi Arabia (3.75 riyals per USD), the UAE (3.6725 dirhams), Qatar (3.64 riyals), Bahrain, Oman and Jordan — generally move their policy rates in line with the US Federal Reserve to maintain the peg, because free capital movement and a fixed exchange rate leave little room for independent monetary policy (the "impossible trinity"). Kuwait pegs to an undisclosed basket. For companies in these markets, US monetary policy largely determines local borrowing costs. Verify current regimes and rates with the relevant central bank.

37.3 Interest rates and company decisions

WORKED EXAMPLE · A 200 bps rise in rates (fictional company)

A company has $50m of floating-rate debt at reference rate + 2.5%, EBITDA $20m, and is evaluating a project with IRR 11%. - **Interest cost:** reference rate rises from 3% to 5% → all-in from 5.5% to 7.5% →

interest +$1.0m per year; interest cover (EBITDA basis) falls from 7.3× to 5.3×. - **Hurdle rate:** WACC rises perhaps from 9% to 10.5%; the 11% project's NPV shrinks toward zero. - **Customers:** consumers with variable-rate mortgages and businesses with floating debt spend less; demand slows. - **Valuation:** a company valued at 12× EBITDA may be re-rated to 10× as discount rates rise — reducing the value of equity, and the collateral value of shares for lenders. - **Working capital:** carrying inventory and receivables costs more; pressure to shorten terms rises across the supply chain. - **Response:** hedge part of the floating exposure with an interest-rate swap or cap, prioritise high-return projects, accelerate working-capital release.

37.4 The yield curve

The **yield curve** plots government bond yields across maturities. Normally upward-sloping (investors demand more for longer maturities). An **inverted** curve (short rates above long rates) has historically often preceded recessions in the US, as markets expect future rate cuts. For corporates, the curve shape affects the choice between floating and fixed-rate debt and between short and long maturities.

37.5 Fiscal policy and government debt

Governments spend (salaries, subsidies, infrastructure, social programmes), tax, and borrow to fund deficits. Fiscal policy affects businesses directly through government contracts, taxes and subsidies, and indirectly through demand and interest rates.

Indicator	Why business cares
Budget balance (% of GDP)	Large deficits may lead to tax increases or spending cuts; financing pressure
Government debt (% of GDP)	Sustainability; crowding out of private credit; sovereign rating
Interest burden (% of revenue)	Fiscal space for projects
Fiscal breakeven oil price (oil exporters)	Oil price needed to balance the budget; below it, spending and project awards may be cut
Government arrears to contractors	Direct cash-flow risk for contractors and suppliers

WORKED EXAMPLE · Oil price and government spending (fictional country)

An oil exporter's budget assumes oil at $75 per barrel; its fiscal breakeven is $80. If oil averages $60, revenue falls by roughly 20% of the oil share; the government delays new projects and payments to contractors stretch from 90 to 180 days. A contractor with $40m of receivables from government entities needs an extra $20m of working-capital funding — exactly when banks become more cautious about the sector.

37.6 Money, credit and banks

Most money is created by commercial banks when they lend (a loan creates a deposit). Credit growth fuels expansions; credit crunches deepen recessions. Business indicators: private-sector credit growth, loan-to-deposit ratios, interbank rates and spreads, and central-bank liquidity operations.

PRACTICE · Practice questions

1. A company has $80m of debt: $30m fixed at 6% and $50m floating at reference + 2%. The reference rate falls from 4.5% to 3.5%. What is the change in annual interest?
2. Why do many Gulf central banks follow the US Federal Reserve's rate decisions?

3. An importer finances inventory with an overdraft at reference + 3%. Rates rise 150 bps; average overdraft $6m; gross margin on goods 18%; inventory days 90. Estimate the margin impact as a % of sales.

ANSWER · Answers to the practice questions

1. Only the floating part changes: $50m × −1.0% = **−$500,000** per year.
2. Their currencies are pegged to the US dollar; with open capital accounts, maintaining the peg requires interest rates aligned with US rates, otherwise capital would flow out (if local rates were too low) or in (if too high), pressuring the peg.
3. Extra interest = $6m × 1.5% = $90,000 per year. Annual sales: inventory $6m at 90 days of COGS → COGS ≈ $24.3m → sales ≈ 24.3 ÷ 0.82 ≈ $29.7m. Impact ≈ 90 ÷ 29,700 ≈ **0.3% of sales** — about 1.7% of gross profit.

CHAPTER 38

Exchange Rates, the Balance of Payments and International Trade Policy

LEARNING OBJECTIVES · In this chapter

What drives exchange rates, how currency regimes work, the balance of payments, trade deficits and surpluses, capital flows, tariffs and non-tariff barriers — with numerical effects on importers and exporters. FX market mechanics and hedging are in Part XVIII.

38.1 Exchange-rate regimes

Regime	Description	Examples (illustrative)	Business implication
Free float	Market-determined	USD, EUR, GBP, JPY	Volatility; hedging markets deep
Managed float	Market with central-bank intervention	Many emerging markets	Periodic adjustments; intervention risk
Crawling peg / band	Gradual adjustments	Some emerging markets	Predictable depreciation
Hard peg	Fixed to anchor currency	GCC currencies to USD; Jordanian dinar	Low volatility while credible; risk of sudden devaluation if reserves fall
Currency board / dollarisation	Domestic money fully backed or replaced	Hong Kong dollar (currency board); Panama (uses USD)	Minimal FX risk vs anchor
Multiple rates / controls	Official and parallel rates; restricted access	Countries in crisis	Trapped cash; pricing complexity

COMMON MISTAKES · Peg risk is real

Pegs can last decades and then fail abruptly when reserves run out. Lebanon's long-standing peg of about 1,507.5 pounds per dollar collapsed after the 2019 financial crisis; Egypt devalued sharply in 2016 and again in 2022–2024. Companies with local-currency revenue and USD costs or debt suffered severe losses. Assess the sustainability of a peg: FX reserves (months of imports), current-account balance, external debt, fiscal position, and existence of a parallel market rate.

38.2 What moves exchange rates

- **Interest-rate differentials** and expected monetary policy (capital seeks higher risk-adjusted returns).
- **Inflation differentials** (over the long run, **purchasing power parity**: high-inflation currencies tend to depreciate).
- **Current-account balance** and terms of trade (commodity exporters' currencies often track commodity prices).
- **Capital flows** and risk sentiment ("risk-on/risk-off"; safe-haven currencies).
- **Political and policy uncertainty**, sanctions, capital controls.
- **Central-bank intervention.**

FORMULA · Parity relationships (approximations)

Relative PPP: expected % change in exchange rate ≈ inflation(domestic) − inflation(foreign)
Covered interest parity: Forward = Spot × (1 + i_quote) ÷ (1 + i_base) (for the period) — this one holds closely in practice because of arbitrage

38.3 The balance of payments

Account	Components	Interpretation
Current account	Trade in goods, trade in services, primary income (interest, dividends, wages), secondary income (remittances, aid)	Deficit = country spends more abroad than it earns; must be financed
Capital account	Capital transfers, non-produced assets	Usually small
Financial account	Foreign direct investment, portfolio flows, loans, changes in reserves	Financing of the current account

A **trade deficit** is not automatically bad (a fast-growing economy importing capital goods may run one), but deficits financed by short-term, volatile inflows create vulnerability. **Remittances** are a major current-account inflow for countries such as Egypt, Lebanon and Jordan; **hydrocarbon exports** dominate Gulf current accounts.

38.4 Currency moves and company results

WORKED EXAMPLE · Importer and exporter hit by a 20% depreciation (fictional)

A local currency (LCY) depreciates from 10 to 12 per USD (the USD costs 20% more; the LCY has lost 16.7% of its USD value).

Importer sells in LCY, buys in USD. Annual imports $5m; sales LCY 70m; other costs LCY 10m. - Before: COGS = 5m × 10 = LCY 50m → profit = 70 − 50 − 10 = **LCY 10m**. - After (prices unchanged): COGS = LCY 60m → profit **LCY 0**. To restore profit it must raise prices by about 14% — possible only if competitors face the same cost and demand allows.

Exporter sells in USD, costs in LCY. Exports $8m; costs LCY 70m. - Before: revenue LCY 80m → profit **LCY 10m**. - After: revenue LCY 96m → profit **LCY 26m** (+160%), unless it imports inputs priced in USD.

Borrower with USD debt of $10m and LCY revenue: debt rises from LCY 100m to LCY 120m; leverage ratios jump overnight.

38.5 Tariffs and trade barriers

Barrier	Description	Business effect
Ad valorem tariff	% of customs value	Raises landed cost
Specific duty	Fixed amount per unit/weight	Hits low-value goods proportionally more
Anti-dumping / countervailing duties	Additional duties on unfairly priced or subsidised imports	Sudden large cost increases on specific origins
Quotas	Quantity limits	Scarcity; licence value
Import licences / registrations	Permission required	Delays; market access barrier
Technical standards / conformity	Product certification, labelling	Testing costs; delays

Barrier	Description	Business effect
Local content requirements	Minimum local value in government contracts	Need for local partners/production
Export restrictions	Bans or taxes on exports	Supply shocks for buyers abroad
Rules of origin	Determine eligibility for preferential tariffs	Sourcing and documentation

Free trade agreements and customs unions (e.g., the GCC Customs Union, the Greater Arab Free Trade Area, EU agreements with Mediterranean partners) can reduce duties for goods meeting rules of origin — requiring correct certificates of origin.

WORKED EXAMPLE · Tariff effect on landed cost and sourcing

A product costs $100 CIF from Country A (MFN duty 12%) or $106 CIF from Country B (FTA partner, 0% duty with valid origin certificate). - From A: duty $12 → **$112**; from B: **$106**. Country B wins despite a higher price — provided the goods qualify under the rules of origin and the certificate is correctly issued.

PRACTICE · Practice questions

1. Inflation is 15% locally and 3% in the US. Under relative PPP, what annual depreciation would you expect?
2. Spot EUR/USD 1.0800; 1-year USD rate 4.5%; EUR rate 2.5%. Compute the 1-year forward.
3. An exporter earns $20m, with USD-priced imported inputs of $6m and local costs of LCY 100m. LCY moves from 8 to 9 per USD. Compute profit before and after in LCY.

ANSWER · Answers to the practice questions

1. Approximately **12%** per year (1.15 ÷ 1.03 − 1 = 11.65%).
2. Forward = 1.0800 × 1.045 ÷ 1.025 = **1.1011** (EUR at a forward premium because EUR rates are lower).
3. Before: revenue 160 − inputs 48 − local 100 = **LCY 12m**. After: 180 − 54 − 100 = **LCY 26m**. The natural hedge (USD inputs) reduces, but does not eliminate, the gain.

CHAPTER 39

Geopolitical, Sanctions, Country and Sovereign Risk — and How Macro Shocks Reach Companies

LEARNING OBJECTIVES · In this chapter

How political and geopolitical events, sanctions and sovereign stress affect commercial transactions; how professionals assess country risk; and a framework that links macroeconomic shocks to company financial statements.

39.1 Country risk and sovereign risk

Country risk is the risk that conditions in a country — political, economic, legal, social — cause losses for businesses and investors there or trading with it. **Sovereign risk** is the risk that a government defaults on its own obligations, or uses its powers to stop others paying (transfer and convertibility restrictions, moratoria, expropriation).

Component	Examples	Indicators
Political	Instability, regime change, conflict, corruption	Governance indicators, election calendars, conflict monitoring
Economic	Recession, inflation, devaluation	GDP, CPI, reserves, current account
Transfer & convertibility	Inability to convert local currency or transfer abroad	Parallel-market premium, central-bank FX allocations, arrears
Legal / regulatory	Weak contract enforcement, sudden rule changes	Rule-of-law indices, arbitration enforcement record
Expropriation	Nationalisation, forced contract renegotiation	History, investment treaties
Sovereign credit	Government default	Credit ratings, CDS spreads, debt/GDP, bond yields

Sources commonly used: sovereign ratings from S&P Global, Moody's and Fitch; OECD country risk classifications for officially supported export credits; export-credit agency country cover policies; IMF Article IV reports and World Bank data; trade credit insurers' country ratings.

39.2 Sanctions

Sanctions regimes (UN, US, EU, UK and others) restrict dealings with designated countries, sectors, entities, individuals and vessels. Key business consequences:

- **Prohibited transactions:** goods, services, financing or payments involving sanctioned parties.
- **Ownership rules:** under some regimes (e.g., US OFAC's 50% rule), entities owned 50% or more by blocked persons are treated as blocked even if not listed.
- **USD nexus:** payments in USD clearing through US banks can bring transactions within US jurisdiction.
- **Secondary sanctions:** the US may penalise non-US persons for certain dealings with sanctioned targets.
- **Bank de-risking:** banks may refuse legal transactions involving high-risk jurisdictions, goods or routes.
- **Shipping:** vessel ownership, flag, AIS gaps, ship-to-ship transfers and port calls are screened.

COMMON MISTAKES · Practical sanctions controls for a trading company

Screen all counterparties (including owners and banks), vessels and ports; obtain end-user declarations for sensitive goods; include sanctions clauses in contracts (right to suspend/terminate); check export-control classifications for dual-use items; keep records; escalate red flags (unusual routing, reluctance to identify end users, payments from third parties). Get specialised legal advice — sanctions rules change frequently.

39.3 Geopolitical risk in supply chains

Recent years have shown how conflicts and chokepoints disrupt trade: attacks on shipping in the Red Sea from late 2023 forced many carriers to re-route around the Cape of Good Hope, adding roughly 10–14 days to many Asia–Europe voyages and raising freight and insurance costs; the war in Ukraine disrupted grain, fertiliser and energy flows; pandemic-era port congestion caused record freight rates. Companies respond with diversified suppliers and routes, larger buffer inventories, regional sourcing ("nearshoring"), war-risk insurance, and contract clauses addressing re-routing costs.

39.4 A framework: how macro shocks reach the financial statements

Shock	Income statement	Balance sheet	Cash flow	Who is hurt / helped
Interest rates +200 bps	Higher interest; lower demand from rate-sensitive customers	Lower asset valuations; pension and lease discounting changes	Higher interest paid	Hurt: leveraged firms, real estate, consumers with debt. Helped: cash-rich firms, banks' margins (initially)
Inflation +5 pts	Costs up; revenue up if pricing power; margin squeeze on fixed contracts	Inventory and receivables cost more to carry	Working capital rises in nominal terms	Hurt: fixed-price sellers, lenders at fixed rates. Helped: borrowers at fixed rates, firms with pricing power
Local currency −20%	Importers' costs up; exporters' revenue up	FX debt up in local terms	Hedging settlements	Hurt: importers, FX borrowers. Helped: exporters, tourism
Recession	Volumes down; bad debts up	Impairments	Lower CFO; tighter credit	Hurt: cyclicals. Helped: discounters
Oil price −30%	Lower fuel costs; in oil economies, lower government spending	Upstream asset impairments	Lower revenue for producers	Hurt: producers, oil-economy contractors. Helped: airlines, shipping, importers
Sanctions on a market	Lost sales; stranded receivables	Write-offs of receivables and assets	Trapped cash	Hurt: exposed exporters and banks
Tariff +15%	Landed cost up	Inventory cost up	More working capital	Hurt: importers; helped: domestic producers

بالعربية — كيف تصل الصدمات الاقتصادية الكلية إلى الشركات؟

أي تغيّر في أسعار الفائدة أو التضخم أو سعر الصرف أو أسعار النفط ينتقل إلى قوائم الشركة عبر قنوات محددة: الإيرادات (الطلب والأسعار)، والتكاليف (المدخلات المستوردة والأجور)، وتكلفة التمويل، وقيمة الأصول والالتزامات بالعملات الأجنبية، ورأس المال العامل. المهني المتمرّس يحدد هذه القنوات مسبقاً لكل نشاط، ويقيس حساسية الأرباح والنقد لكل متغير، ثم يتخذ إجراءات للتحوط أو التنويع أو تعديل العقود.

39.5 Building a macro sensitivity for your business

1. List the macro variables that matter (e.g., USD/LCY, policy rate, oil price, construction output, consumer spending).
2. For each, estimate the transmission: revenue elasticity, cost share in foreign currency, floating-rate debt, customer sensitivity.
3. Compute the effect of a plausible shock on EBITDA, net income, cash and covenants.
4. Prioritise by impact and probability; define hedges, contract changes and early-warning indicators.

WORKED EXAMPLE · Macro sensitivity table (fictional importer-distributor)

Shock	Δ EBITDA ($m)	Δ Net income ($m)	Covenant headroom (net debt/EBITDA, limit 3.0×; base 2.2×)
USD/LCY +15%, 50% pass-through to prices after 3 months	−2.1	−2.4	2.9× (tight)
Rates +200 bps on $25m floating debt	0	−0.4	2.2× (EBITDA unaffected; interest cover falls)
Demand −10%	−1.8	−1.5	2.7×
Combined (all three)	−3.9	−4.3	3.6× (breach)

Conclusion: the combined downside breaches the covenant. Actions: hedge 60% of forecast USD purchases 6 months forward; fix half of the floating debt with a swap; negotiate a covenant at 3.5× in the next refinancing; introduce price-adjustment clauses with key customers.

PRACTICE · Practice questions

1. Name four indicators you would review before granting 120-day open-account terms to a buyer in a country with a currency peg under pressure.
2. Your company sells to a state-owned buyer in a country whose government is in arrears with contractors. Which instruments could protect you?
3. Classify each as country, sovereign or counterparty risk: (a) the central bank stops allocating USD for imports; (b) the buyer goes bankrupt; (c) the government defaults on its eurobonds; (d) new import licensing rules block your goods at the port.

ANSWER · Answers to the practice questions

1. FX reserves in months of imports; parallel-market premium vs official rate; current-account balance and external debt service; central-bank FX allocation backlogs and payment delays; sovereign rating/CDS trend; IMF programme status.
2. Confirmed L/C (from a strong foreign bank), credit insurance including political risk (transfer/convertibility), export-credit agency cover, advance payments, escrow, or payment in an offshore account.
3. (a) Country (transfer and convertibility); (b) counterparty (commercial credit); (c) sovereign; (d) country (regulatory/legal).

SUMMARY · Part V summary

- Supply, demand and elasticity explain price moves and pricing power; market structure explains margins.
- GDP composition, inflation and the business cycle drive demand and costs; inflation punishes fixed-price commitments without indexation.

- Interest rates transmit through financing costs, demand, valuations and working capital; pegs import the anchor's monetary policy.
- Exchange-rate changes redistribute profits between importers and exporters and can overwhelm borrowers with FX debt.
- Tariffs, sanctions and geopolitical shocks require structured screening, diversification and contract protection.
- Professionals translate macro scenarios into effects on EBITDA, cash and covenants — and act before the shock.

PART XX

Business Mathematics and Commercial Calculations

رياضيات الأعمال والحسابات التجارية

This part is a working reference for the calculations that business professionals do every day — in quotations, credit decisions, shipment costing, loan negotiations, budget reviews and investment memos. Each calculation card follows the same seven elements: the formula, a plain-English explanation, an Arabic explanation, a simple example, a realistic business example, a worked solution, and common mistakes.

Use it as a workbook: cover the solution, compute it yourself (on paper, in a spreadsheet, or in Python), then check. Speed and accuracy in these calculations are part of professional credibility — in a negotiation, the person who can compute the effect of a 2% discount on margin in ten seconds controls the conversation.

LEARNING OBJECTIVES · Learning objectives for Part XX

- Perform percentage, margin, markup, discount, commission and pricing calculations without error.
- Compute simple and compound interest, effective rates, loan payments and amortisation schedules.
- Convert currencies using bid/ask quotes, cross rates and forward points.
- Calculate profitability, break-even, working-capital and ratio measures.
- Compute landed cost and allocate freight; calculate investment returns, NPV, valuation multiples, expected values and simple forecasts.

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CHAPTER 40

Commercial Arithmetic: Percentages, Margins, Discounts, Commissions and Pricing

40.1 Percentages and percentage points

Percentage change	
FORMULA	**% change = (New − Old) ÷ Old × 100**
PLAIN ENGLISH	How much something grew or shrank relative to where it started. A change in a rate is expressed in percentage points (pp); the relative change of that rate is a percentage.
بالعربية	نسبة التغير = (القيمة الجديدة − القيمة القديمة) ÷ القيمة القديمة. أما التغير في نسبة مئوية (مثل الهامش من 20% إلى 22%) فيُعبَّر عنه بالنقاط المئوية (2 نقطة مئوية)، وهو يعادل زيادة نسبية قدرها 10%.
SIMPLE EXAMPLE	Price rises from $80 to $100: (100 − 80) ÷ 80 = 25%. Falling back from $100 to $80 is −20%, not −25%.
BUSINESS EXAMPLE	Gross margin improved from 20% to 22% while revenue grew from $40m to $46m. Describe both changes correctly.
WORKED SOLUTION	Revenue growth = (46 − 40) ÷ 40 = **15%**. Margin change = **+2 percentage points**, which is a **10% relative improvement** (2 ÷ 20). Gross profit grew from $8.0m to $10.12m = **+26.5%**.
COMMON MISTAKES	Saying “margin rose 2%” when it rose 2 percentage points; assuming a 20% fall followed by a 20% rise returns to the start (0.8 × 1.2 = 0.96); averaging percentages of different bases.

Reverse percentage (tax-inclusive amounts)	
FORMULA	**Net amount = Gross amount ÷ (1 + rate)** **Tax = Gross − Net = Gross × rate ÷ (1 + rate)**
PLAIN ENGLISH	When a price includes tax (VAT, GST), you cannot take the tax rate of the gross amount to find the tax; divide by one plus the rate.
بالعربية	إذا كان المبلغ شاملاً للضريبة فإن صافي المبلغ = المبلغ الإجمالي ÷ (1 + نسبة الضريبة). الخطأ الشائع هو حساب نسبة الضريبة من المبلغ الإجمالي مباشرة.
SIMPLE EXAMPLE	$115 including 15% VAT → net $100, VAT $15 (not 15% × 115 = $17.25).
BUSINESS EXAMPLE	A UAE supplier invoice shows AED 52,500 including 5% VAT. How much is input VAT, and what is the cost if VAT is recoverable?
WORKED SOLUTION	Net = 52,500 ÷ 1.05 = **AED 50,000**; VAT = **AED 2,500**. If the buyer is VAT-registered and can recover input VAT, the cost is AED 50,000; the AED 2,500 is a temporary cash outflow recovered through the VAT return.
COMMON MISTAKES	Computing VAT as 5% × 52,500 = 2,625; treating recoverable VAT as a cost in pricing; forgetting that VAT is a cash-flow timing item for registered businesses.

40.2 Margins, markups and pricing

Gross margin and markup

FORMULA	**Margin = (Price − Cost) ÷ Price** **Markup = (Price − Cost) ÷ Cost** **Margin = Markup ÷ (1 + Markup)** **Markup = Margin ÷ (1 − Margin)**
PLAIN ENGLISH	Margin is profit as a share of the selling price; markup is profit as a share of cost. The same deal always has a higher markup than margin.
بالعربية	الهامش = الربح ÷ سعر البيع، ونسبة الإضافة على التكلفة = الربح ÷ التكلفة. للصفقة نفسها تكون نسبة الإضافة دائماً أعلى من الهامش. مثلاً: إضافة 25% على التكلفة تعطي هامشاً قدره 20%.
SIMPLE EXAMPLE	Cost $75, price $100: margin 25%, markup 33.3%.
BUSINESS EXAMPLE	A distributor's policy is a 35% gross margin. The sales team uses a 35% markup on landed cost. Product landed cost is $64. What price does each method give, and what margin does the sales team actually achieve?
WORKED SOLUTION	Correct price = 64 ÷ (1 − 0.35) = **$98.46**. Sales team price = 64 × 1.35 = **$86.40** → margin = 22.40 ÷ 86.40 = **25.9%**, nearly 9 percentage points below policy.
COMMON MISTAKES	Using markup and margin interchangeably; computing price as cost × (1 + target margin); calculating margin on cost excluding freight and duties.

Price from a target margin

FORMULA	**Price = Cost ÷ (1 − Target margin)**
PLAIN ENGLISH	To achieve a margin m on the selling price, divide cost by (1 − m).
بالعربية	للوصول إلى هامش مستهدف على سعر البيع، يُقسم سعر التكلفة على (1 − الهامش المستهدف).
SIMPLE EXAMPLE	Cost $60, target margin 40% → price = 60 ÷ 0.6 = $100.
BUSINESS EXAMPLE	A trader's landed cost is $11.72 per unit. Commission to an agent is 3% of the selling price and the target net margin after commission is 25%. Find the price.
WORKED SOLUTION	Price must cover cost plus 28% of price (25% margin + 3% commission): P = 11.72 ÷ (1 − 0.28) = **$16.28**. Check: commission 0.49; margin 16.28 − 11.72 − 0.49 = 4.07 = 25.0% of price.
COMMON MISTAKES	Adding percentages of price to cost (11.72 × 1.28 = 15.00 → only 18.9% margin after commission).

40.3 Discounts and commissions

Chain (successive) discounts

FORMULA	**Net price = List × $(1 - d_1) \times (1 - d_2) \times \ldots \times (1 - d_n)$** **Effective discount = $1 - \Pi(1 - d_i)$**
PLAIN ENGLISH	Discounts applied one after another do not add up; each applies to the already reduced price.
بالعربية	الخصومات المتتالية لا تُجمع، لأن كل خصم يُطبق على السعر بعد الخصم السابق. خصم 20% ثم 10% يساوي خصماً فعلياً قدره 28% وليس 30%.
SIMPLE EXAMPLE	20% then 10%: 0.8 × 0.9 = 0.72 → effective discount 28%.
BUSINESS EXAMPLE	List price $2,500; trade discount 25%, quantity discount 5%, prompt-payment discount 2%. Net price and effective discount?
WORKED SOLUTION	2,500 × 0.75 × 0.95 × 0.98 = **$1,745.63**; effective discount = 1 − 0.69825 = **30.2%**.
COMMON MISTAKES	Adding discounts (25 + 5 + 2 = 32%); forgetting that prompt-payment discounts depend on payment timing and may not be taken.

Commission (flat, tiered, margin-based)

FORMULA	**Flat: Commission = Rate × Base** **Tiered: Σ (rate of tier × amount within tier)** **Margin-based: Rate × Gross profit**
PLAIN ENGLISH	Commission rewards salespeople or agents; the base (revenue or margin) and the tier structure shape their behaviour.
بالعربية	العمولة مكافأة للبائع أو الوكيل، وتُحسب على أساس المبيعات أو هامش الربح، وقد تكون بشرائح متدرجة. حساب العمولة على المبيعات قد يدفع البائع إلى منح خصومات كبيرة، أما حسابها على الهامش فيربط مصلحته بربحية الشركة.
SIMPLE EXAMPLE	4% of $50,000 sales = $2,000.
BUSINESS EXAMPLE	Agent earns 3% on sales up to $1m and 5% above, per year. Sales $1.6m. Alternatively the company proposes 18% of gross profit; gross margin on these sales is 24%. Compare.
WORKED SOLUTION	Tiered: 3% × 1,000,000 + 5% × 600,000 = 30,000 + 30,000 = **$60,000**. Margin-based: 18% × (24% × 1.6m) = 18% × 384,000 = **$69,120**. The margin-based plan pays more at this margin but would pay less if the agent sold at heavy discounts — aligning incentives.
COMMON MISTAKES	Applying the higher tier rate to all sales; paying commission on invoiced but uncollected sales without clawback; ignoring that commissions on discounted deals reduce margin twice.

Blended (weighted average) margin and mix effect

FORMULA	**Blended margin = Σ (share of revenue$_i$ × margin$_i$)**
PLAIN ENGLISH	The overall margin is a weighted average; a shift in sales mix changes it even if each product's margin is unchanged.
بالعربية	الهامش الإجمالي متوسط مرجّح لهوامش المنتجات بحسب حصة كل منها من الإيرادات. تغيّر مزيج المبيعات يغيّر الهامش الإجمالي حتى لو بقيت هوامش المنتجات ثابتة.
SIMPLE EXAMPLE	A: 60% of sales at 40%; B: 40% at 20% → 0.6 × 40 + 0.4 × 20 = 32%.
BUSINESS EXAMPLE	Next year the mix moves to 45% A / 55% B with unchanged product margins. Revenue rises 10% from $10m. What happens to gross profit?
WORKED SOLUTION	New blended margin = 0.45 × 40 + 0.55 × 20 = **29%**. Gross profit = 11m × 29% = **$3.19m** vs $3.20m this year — revenue +10%, gross profit flat. This is a **mix effect**.
COMMON MISTAKES	Averaging margins without weights; attributing margin decline to "pricing" when it is mix.

PRACTICE · Practice questions

1. Cost $42, target margin 30%. Price? What markup is that?
2. List price $800; discounts 15% and 10%; then 5% VAT on the net. Total invoice?
3. A price was cut 10% and volume rose 12%. Revenue change?
4. Margin fell from 18.0% to 16.5%. Express the change in pp and in relative terms.

ANSWER · Answers to the practice questions

1. 42 ÷ 0.70 = **$60.00**; markup = 18 ÷ 42 = **42.9%**.
2. 800 × 0.85 × 0.90 = 612.00; VAT 30.60 → **$642.60**.
3. 0.90 × 1.12 = 1.008 → **+0.8%**.
4. **−1.5 pp**; relative change = −1.5 ÷ 18 = **−8.3%**.

CHAPTER 41

Interest, Loans and Foreign Exchange Calculations

41.1 Interest

Simple interest with day count

FORMULA	**Interest = Principal × Annual rate × (Days ÷ Day basis)**
PLAIN ENGLISH	Short-term loans, trade finance, deposits and money-market instruments usually pay simple interest for the period, calculated on actual days over a 360- or 365-day year depending on currency and contract.
بالعربية	الفائدة البسيطة = أصل المبلغ × سعر الفائدة السنوي × (عدد الأيام ÷ أساس السنة). يُستخدم أساس 360 يوماً للدولار واليورو في الغالب، و365 يوماً للجنيه الإسترليني وبعض العملات المحلية؛ يجب الرجوع إلى العقد.
SIMPLE EXAMPLE	$10,000 at 6% for 90 days (act/360) = 10,000 × 0.06 × 90 ÷ 360 = $150.
BUSINESS EXAMPLE	A bank provides a $2m import loan for 120 days at 3M SOFR 4.30% + 2.95% margin (act/360). Interest due at maturity?
WORKED SOLUTION	Rate 7.25%. Interest = 2,000,000 × 0.0725 × 120 ÷ 360 = **$48,333.33**. (On act/365 it would be $47,671.23.)
COMMON MISTAKES	Using 365 for USD loans documented on act/360; forgetting fees (arrangement, L/C commission) in the all-in cost.

Compound growth and CAGR

FORMULA	**$FV = PV \times (1 + r)^n$** **CAGR = (End value ÷ Start value)^(1/n) − 1**
PLAIN ENGLISH	Compound growth applies a rate to an amount that already includes previous growth. CAGR is the constant annual rate that turns the start value into the end value.
بالعربية	النمو المركب يطبّق المعدل على مبلغ يشمل النمو السابق. ومعدل النمو السنوي المركب (CAGR) هو المعدل الثابت الذي يحوّل القيمة الأولى إلى القيمة الأخيرة خلال عدد السنوات.
SIMPLE EXAMPLE	$1,000 at 10% for 3 years = 1,000 × 1.331 = $1,331.
BUSINESS EXAMPLE	Revenue grew from $12.0m to $19.3m over 4 years. What is the CAGR? What would revenue be in 2 more years at the same rate?
WORKED SOLUTION	CAGR = (19.3 ÷ 12.0)^(1/4) − 1 = 1.6083^0.25 − 1 = **12.6%**. In 2 years: 19.3×1.126^2 = **$24.5m**.
COMMON MISTAKES	Dividing total growth by years (60.8% ÷ 4 = 15.2% overstates); counting the number of data points instead of periods.

Flat rate vs effective rate

FORMULA	**Flat interest = Principal × Flat rate × Years (charged on the original principal). Effective rate: solve Principal = Payment × [1 − (1 + r)⁻ⁿ] ÷ r for r.**
PLAIN ENGLISH	Flat-rate quotes charge interest on the full original principal even though it is repaid gradually, so the true (effective) rate is much higher.
بالعربية	الفائدة الثابتة (Flat) تُحتسب على أصل القرض الأصلي طوال المدة رغم سداده تدريجياً، لذلك يكون المعدل الفعلي أعلى بكثير — قرابة الضعف تقريباً في القروض المقسطة.
SIMPLE EXAMPLE	$10,000 for 1 year at 10% flat repaid monthly: interest $1,000; effective APR ≈ 18%.
BUSINESS EXAMPLE	A vehicle loan of $30,000 for 3 years at 8% flat, monthly instalments. Compute the instalment and approximate effective rate.
WORKED SOLUTION	Interest = 30,000 × 8% × 3 = $7,200 → instalment = 37,200 ÷ 36 = **$1,033.33**. Solving for r: ≈ 1.21% per month → nominal APR ≈ **14.6%**, EAR ≈ **15.6%** — almost double the quoted 8%.
COMMON MISTAKES	Comparing a flat-rate quote with a reducing-balance rate as if they were equivalent.

41.2 Loans

Loan payment and amortisation schedule

FORMULA	**PMT = P × r ÷ [1 − (1 + r)⁻ⁿ]; each period: Interest = Opening balance × r; Principal = PMT − Interest; Closing = Opening − Principal**
PLAIN ENGLISH	An annuity loan has equal payments; early payments are mostly interest, later ones mostly principal.
بالعربية	في القرض ذي الأقساط المتساوية تكون الدفعات الأولى معظمها فوائد والأخيرة معظمها سداد لأصل الدين. يُحسب القسط بالمعادلة أعلاه، ثم يُعد جدول السداد فترةً بعد فترة.
SIMPLE EXAMPLE	$10,000 over 2 years, annual payments, 10%: PMT = 10,000 × 0.1 ÷ (1 − 1.1^{-2}) = $5,761.90.
BUSINESS EXAMPLE	Equipment loan $240,000, 2 years, quarterly payments, 8% p.a. (2% per quarter). Compute the payment and the first three lines of the schedule. Compare with equal-principal repayment.
WORKED SOLUTION	PMT = 240,000 × 0.02 ÷ (1 − 1.02^{-8}) = 4,800 ÷ 0.146510 = **$32,762.36**.

QUARTER	OPENING	INTEREST	PRINCIPAL	CLOSING
1	240,000.00	4,800.00	27,962.36	212,037.64
2	212,037.64	4,240.75	28,521.61	183,516.03
3	183,516.03	3,670.32	29,092.04	154,423.99

Equal principal: $30,000 principal per quarter + interest (Q1 payment $34,800, falling to $30,600 in Q8); total interest lower ($21,600 vs $22,098.88) because principal is repaid faster early on.

COMMON MISTAKES	Treating the whole instalment as an expense (only interest is); using the annual rate with quarterly periods; forgetting fees and balloon payments.

41.3 Foreign exchange

Currency conversion and bid/ask

FORMULA	**Quote "BASE/QUOTE = rate" means 1 BASE = rate × QUOTE. Bank buys base at the bid, sells base at the ask (offer). Quote amount = Base amount × rate; Base amount = Quote amount ÷ rate.**
PLAIN ENGLISH	Always identify the base currency. You buy the base currency at the higher (ask) price and sell it at the lower (bid) price; the difference is the bank's spread.
بالعربية	في التسعيرة EUR/USD = 1.0850، العملة الأساسية هي اليورو، أي أن اليورو الواحد يساوي 1.0850 دولار. يشتري البنك العملة الأساسية بسعر الشراء (Bid) ويبيعها بسعر البيع (Ask) الأعلى، والفرق هو هامش البنك.
SIMPLE EXAMPLE	USD/JPY 150.00: ¥1,500,000 = $10,000.
BUSINESS EXAMPLE	EUR/USD bid 1.0848, ask 1.0852. An importer must pay EUR 250,000; an exporter receives EUR 250,000. What USD amounts apply?
WORKED SOLUTION	Importer buys EUR (base) at the ask: 250,000 × 1.0852 = **$271,300**. Exporter sells EUR at the bid: 250,000 × 1.0848 = **$271,200**. The $100 difference on each side is the spread (corporate spreads on smaller amounts are often wider).
COMMON MISTAKES	Inverting the quote (dividing instead of multiplying); using the mid-rate for budgeting without the spread; ignoring bank charges on transfers.

Cross rates

FORMULA	**A/C = A/B × B/C; if both quoted against USD: (A/USD) = (A/USD), (USD/C) → A/C = A/USD × USD/C; for two USD-base quotes: $C_1/C_2 = (USD/C_2) \div (USD/C_1)$**
PLAIN ENGLISH	When two currencies are not directly quoted, derive their rate through a common currency (usually USD), making sure the units cancel.
بالعربية	عندما لا تتوافر تسعيرة مباشرة بين عملتين، نحسب السعر عبر عملة وسيطة (غالباً الدولار) مع التأكد من أن الوحدات تُلغي بعضها بشكل صحيح.
SIMPLE EXAMPLE	EUR/USD 1.0850, USD/AED 3.6725 → EUR/AED = 1.0850 × 3.6725 = 3.9847.
BUSINESS EXAMPLE	A Saudi importer receives a quotation of CNY 480,000. USD/CNY = 7.20 and USD/SAR = 3.75. Cost in SAR?
WORKED SOLUTION	SAR/CNY = 7.20 ÷ 3.75 = 1.92 CNY per SAR → SAR cost = 480,000 ÷ 1.92 = **SAR 250,000** (equivalently 480,000 ÷ 7.20 = $66,666.67 × 3.75).
COMMON MISTAKES	Multiplying when you should divide; ignoring that each leg has its own bid/ask spread.

Forward rate from interest differentials

FORMULA	**Forward = Spot × (1 + i_quote × d/basis) ÷ (1 + i_base × d/basis); Forward points = Forward − Spot**
PLAIN ENGLISH	A forward rate is not a forecast: it is the spot rate adjusted for the interest-rate difference between the two currencies over the period. The higher-interest currency trades at a forward discount.
بالعربية	سعر الصرف الآجل ليس توقعاً لسعر المستقبل، بل هو السعر الفوري معدلاً بفارق أسعار الفائدة بين العملتين خلال المدة. العملة ذات الفائدة الأعلى تُتداول بخصم في السوق الآجلة.
SIMPLE EXAMPLE	Spot 1.0000; quote-currency rate 5%, base 3%, 1 year → forward ≈ 1.0194.
BUSINESS EXAMPLE	EUR/USD spot 1.0850; USD 3-month rate 4.40%; EUR 2.40%; 90 days, act/360 for both. Compute the forward and the points.
WORKED SOLUTION	Forward = 1.0850 × (1 + 0.044 × 0.25) ÷ (1 + 0.024 × 0.25) = 1.0850 × 1.011 ÷ 1.006 = **1.0904**; forward points ≈ **+54 pips** (EUR at a forward premium).
COMMON MISTAKES	Treating forward points as the bank's fee; reading the forward as the expected future spot; using the wrong currency as base.

PRACTICE · Practice questions

1. Interest on GBP 1,500,000 for 61 days at 6.2% (act/365)?
2. Monthly payment on $85,000, 4 years, 9% p.a.?
3. USD/EGP 49.20 and EUR/USD 1.0700: EUR/EGP?
4. An exporter will receive USD 1m in 6 months; USD rate 4.5%, AED rate 4.4% (both act/360); USD/AED spot 3.6725. Compute the 6-month forward.

ANSWER · Answers to the practice questions

1. 1,500,000 × 0.062 × 61 ÷ 365 = **GBP 15,542.47**.
2. r = 0.0075, n = 48: $(1.0075)^{48}$ = 1.4314 → PMT = 85,000 × 0.0075 ÷ (1 − 1/1.4314) = 637.50 ÷ 0.30138 = **≈ $2,115**.
3. EUR/EGP = 1.0700 × 49.20 = **52.64**.
4. Forward = 3.6725 × (1 + 0.044 × 0.5) ÷ (1 + 0.045 × 0.5) = 3.6725 × 1.022 ÷ 1.0225 = **3.6707** (USD at a slight discount; in practice the pegged forward market has its own liquidity and spreads).

CHAPTER 42

Profitability, Working Capital and Financial Ratio Calculations

Contribution margin and multi-product break-even

FORMULA	**CM per unit = Price − Variable cost; CM ratio = CM ÷ Price; Break-even (multi-product) = Fixed costs ÷ Weighted CM per mix unit**
PLAIN ENGLISH	Contribution is what each sale contributes to fixed costs and profit. With several products, break-even depends on the sales mix.
بالعربية	هامش المساهمة هو ما تساهم به كل وحدة مبيعة في تغطية التكاليف الثابتة وتحقيق الربح. وعند تعدد المنتجات تعتمد نقطة التعادل على مزيج المبيعات.
SIMPLE EXAMPLE	Price 50, variable 30 → CM 20, CM ratio 40%; fixed costs 100,000 → break-even 5,000 units.
BUSINESS EXAMPLE	Product A: price 100, CM 40. Product B: price 50, CM 10. Sales mix 1 A : 3 B (in units). Fixed costs $350,000. Break-even units of each?
WORKED SOLUTION	Weighted CM per mix (1A + 3B) = 40 + 30 = 70 → mixes = 350,000 ÷ 70 = 5,000 → **5,000 A and 15,000 B**; break-even revenue = 5,000 × 100 + 15,000 × 50 = **$1.25m**.
COMMON MISTAKES	Using simple average CM (25) instead of the mix-weighted one; including fixed costs allocated per unit in “variable cost”.

Liquidity ratios

FORMULA	**Current ratio = Current assets ÷ Current liabilities; Quick ratio = (Cash + Receivables + Short-term investments) ÷ Current liabilities; Working capital = Current assets − Current liabilities**
PLAIN ENGLISH	Measures of short-term ability to pay. A high ratio built on slow-moving stock or doubtful receivables is not true liquidity.
بالعربية	نسب السيولة تقيس قدرة المنشأة على سداد التزاماتها قصيرة الأجل. لكن نسبة مرتفعة قائمة على مخزون راكد أو ذمم مشكوك في تحصيلها لا تعني سيولة حقيقية.
SIMPLE EXAMPLE	CA 600, CL 400 → current ratio 1.5×; working capital 200.
BUSINESS EXAMPLE	Cash 0.3m; receivables 2.5m; inventory 3.2m; prepayments 0.2m; payables 2.4m; short-term debt 1.6m; current portion of term loan 0.5m. Compute current and quick ratios and comment.
WORKED SOLUTION	CA = 6.2m; CL = 4.5m → current ratio **1.38**×; quick = (0.3 + 2.5) ÷ 4.5 = **0.62**×. Liquidity depends heavily on converting inventory; a lender will examine inventory ageing and the 13-week cash forecast.
COMMON MISTAKES	Including long-term items; ignoring the current portion of long-term debt; treating all inventory as easily saleable.

DSO, DIO, DPO and cash conversion cycle

FORMULA	**DSO = Receivables ÷ Revenue × 365; DIO = Inventory ÷ COGS × 365; DPO = Payables ÷ COGS (or purchases) × 365; CCC = DIO + DSO − DPO; Cash released = Δdays × daily revenue (or daily COGS)**
PLAIN ENGLISH	These convert balance-sheet amounts into days of activity, showing how long cash is tied up in the operating cycle.
بالعربية	هذه المؤشرات تحوّل أرصدة الميزانية إلى عدد أيام من النشاط: أيام تحصيل الذمم المدينة، وأيام بقاء المخزون، وأيام سداد الموردين. ودورة التحويل النقدي = أيام المخزون + أيام التحصيل − أيام السداد.
SIMPLE EXAMPLE	Receivables 100, revenue 1,200 → DSO = 30.4 days.
BUSINESS EXAMPLE	Revenue $48m; COGS $36m; receivables $9.2m; inventory $7.4m; payables $4.9m. Compute the days and CCC; what cash is released if DSO falls to 55 days?
WORKED SOLUTION	DSO = 9.2 ÷ 48 × 365 = **70.0**; DIO = 7.4 ÷ 36 × 365 = **75.0**; DPO = 4.9 ÷ 36 × 365 = **49.7**; CCC = **95.3 days**. DSO to 55: 15 days × 48m ÷ 365 = **$1.97m** released.
COMMON MISTAKES	Using year-end balances when the business is seasonal (use averages); mixing revenue and COGS denominators; comparing DSO across firms with different VAT treatment of receivables.

Leverage and coverage ratios

FORMULA	**Debt/Equity = Total debt ÷ Equity; Net debt/EBITDA = (Debt − Cash) ÷ EBITDA; Interest cover = EBIT (or EBITDA) ÷ Interest; DSCR = (EBITDA − Tax paid − Maintenance capex ± ΔWC) ÷ (Interest + Scheduled principal)**
PLAIN ENGLISH	Measures of the debt burden and the ability to service it from earnings and cash flow. Facility agreements define each precisely.
بالعربية	نسب الرافعة والتغطية تقيس عبء الدين والقدرة على خدمته من الأرباح والتدفقات النقدية. وتحدد اتفاقيات التسهيلات تعريف كل نسبة بدقة.
SIMPLE EXAMPLE	Debt 60, cash 10, EBITDA 25 → net debt/EBITDA 2.0×.
BUSINESS EXAMPLE	EBITDA $18m; EBIT $12m; interest $3m; tax paid $2m; maintenance capex $2.5m; working-capital increase $1.5m; scheduled principal $6m; debt $54m; cash $6m; equity $40m. Compute all four ratios.
WORKED SOLUTION	D/E = **1.35**×; net debt/EBITDA = 48 ÷ 18 = **2.67**×; interest cover (EBIT) = **4.0**× (EBITDA basis 6.0×); DSCR = (18 − 2 − 2.5 − 1.5) ÷ (3 + 6) = 12 ÷ 9 = **1.33**×.
COMMON MISTAKES	Using a different definition from the one in the loan agreement; forgetting leases (post-IFRS 16) or guarantees; using EBITDA instead of cash available for DSCR.

Return ratios	
FORMULA	**ROA = Net income ÷ Avg total assets; ROE = Net income ÷ Avg equity; ROCE = EBIT ÷ (Total assets − Current liabilities); ROIC = EBIT × (1 − t) ÷ (Net debt + Equity)**
PLAIN ENGLISH	Return ratios relate profit to the resources used. ROIC compared with WACC shows whether the business creates value.
بالعربية	نسب العائد تربط الربح بالموارد المستخدمة لتحقيقه. ومقارنة العائد على رأس المال المستثمر (ROIC) بتكلفة رأس المال (WACC) تبيّن ما إذا كانت الشركة تخلق قيمة.
SIMPLE EXAMPLE	NI 10, average equity 50 → ROE 20%.
BUSINESS EXAMPLE	EBIT $12m; tax 20%; net debt $48m; equity $40m; WACC 9.5%. Does the company create value?
WORKED SOLUTION	NOPAT = 9.6m; invested capital 88m → ROIC = **10.9%** > WACC 9.5% → creates value, with a spread of 1.4 percentage points (~$1.2m of economic profit a year).
COMMON MISTAKES	Using closing rather than average balances when they change significantly; comparing ROE across firms with very different leverage.

PRACTICE · Practice questions

1. CM ratio 35%, fixed costs $840,000. Break-even revenue? Revenue for $210,000 profit?
2. Revenue $73m, receivables $14.6m. DSO? If revenue grows 20% at the same DSO, how much more cash is tied up in receivables?
3. EBITDA $10m, debt $38m, cash $3m, covenant 3.25×. How much can EBITDA fall before a breach (debt unchanged)?

ANSWER · Answers to the practice questions

1. BE = 840,000 ÷ 0.35 = **$2.4m**; with profit: 1,050,000 ÷ 0.35 = **$3.0m**.
2. DSO = 14.6 ÷ 73 × 365 = **73 days**. Receivables at +20% revenue: 14.6 × 1.2 = 17.52 → **+$2.92m**.
3. Net debt 35 → EBITDA at breach = 35 ÷ 3.25 = 10.77 > 10 → **already in breach** (3.5×). EBITDA must rise to $10.77m, or net debt fall to $32.5m.

CHAPTER 43

Trade Costs, Investment Returns, Valuation, Probability and Forecasting

43.1 Landed cost and freight

Landed cost per unit	
FORMULA	**Landed cost = Purchase price + Freight + Insurance + Customs duty + Non-recoverable taxes + Port/terminal and clearance charges + Inland transport + Bank/financing costs + Other (inspection, certification) — per unit = total ÷ units**
PLAIN ENGLISH	The full cost to get goods into your warehouse ready for sale. Duty is usually calculated on the customs value (often CIF), and import VAT on the duty-paid value.
بالعربية	التكلفة الواصلة هي التكلفة الكاملة لوصول البضاعة إلى المستودع جاهزة للبيع: سعر الشراء + الشحن + التأمين + الرسوم الجمركية + الضرائب غير المستردة + رسوم الميناء والتخليص + النقل الداخلي + تكاليف التمويل والمصارف. تُحسب الرسوم الجمركية عادةً على القيمة الجمركية (غالباً CIF).
SIMPLE EXAMPLE	Goods $10,000 FOB + freight $1,200 + insurance $50 = CIF $11,250; duty 5% = $562.50; local charges $400 → landed $12,212.50.
BUSINESS EXAMPLE	2,000 units at $18.00 FOB Ningbo; sea freight $2,800 per 40ft container (one container); insurance 0.12% of 110% of CIF (approximate on CIF of FOB+freight); duty 5% of CIF; port and clearance $650; inland trucking $300; L/C and bank charges $420. Landed cost per unit?
WORKED SOLUTION	FOB = 36,000; freight 2,800 → C&F 38,800; insurance = 0.0012 × 1.1 × 38,800 ≈ **$51.22**; CIF ≈ 38,851.22; duty 5% = **1,942.56**; other = 650 + 300 + 420 = 1,370. Landed total ≈ **$42,163.78** → **$21.08 per unit** (17.1% above FOB).
COMMON MISTAKES	Pricing on FOB; forgetting that duty applies to CIF, not FOB, in many countries; omitting bank, demurrage and inspection costs; treating recoverable VAT as a cost.

Chargeable weight and freight allocation

FORMULA	**Air: volumetric weight (kg) = L × W × H (cm) ÷ 6,000; chargeable weight = max(actual, volumetric). Sea LCL: revenue ton (W/M) = max(tonnes, m^3). Allocation of shared costs: by value, weight, volume or units — choose the driver that best reflects consumption.**
PLAIN ENGLISH	Carriers charge for the space cargo occupies as well as its weight. When one shipment carries several products, freight must be allocated to compute product margins.
بالعربية	يحسب الناقل الجوي الوزن الحجمي (الطول × العرض × الارتفاع بالسنتيمتر ÷ 6000) ويُحاسَب على الأكبر بين الوزن الفعلي والحجمي. وفي الشحن البحري الجزئي يُحاسَب على الأكبر بين الأطنان والأمتار المكعبة. وعند شحن منتجات عدة معاً تُوزَّع التكلفة حسب القيمة أو الوزن أو الحجم.
SIMPLE EXAMPLE	A 60 × 50 × 40 cm carton of 18 kg: volumetric = 120,000 ÷ 6,000 = 20 kg → chargeable 20 kg.
BUSINESS EXAMPLE	(a) Air shipment of 10 such cartons at $4.50/kg. (b) LCL shipment: 3.2 tonnes, 7.5 m^3, rate $65 per W/M. (c) A 40ft container freight of $3,000 carries products X ($60,000, 20 m^3), Y ($20,000, 30 m^3) and Z ($20,000, 16 m^3). Allocate by value and by volume.
WORKED SOLUTION	(a) 10 × 20 = 200 kg chargeable × 4.50 = **$900**. (b) 7.5 W/M × 65 = **$487.50**. (c) By value: X **$1,800**, Y **$600**, Z **$600**. By volume (66 m^3): X **$909**, Y **$1,364**, Z **$727**. Bulky low-value Y is much less profitable on a volume basis — usually the more accurate driver for sea freight, while insurance follows value.
COMMON MISTAKES	Using actual weight for light, bulky air cargo; allocating all freight by value (overstates margins of bulky items); forgetting minimum charges.

43.2 Returns and valuation

NPV and IRR (quick reference)

FORMULA	**NPV = Σ $CF_t \div (1 + r)^t$; IRR = r where NPV = 0**
PLAIN ENGLISH	NPV measures value created at a required return; IRR is the break-even return. Full treatment: Chapter 28.
بالعربية	صافي القيمة الحالية يقيس القيمة المضافة عند معدل العائد المطلوب، ومعدل العائد الداخلي هو المعدل الذي تصبح عنده صافي القيمة الحالية صفراً.
SIMPLE EXAMPLE	−100 today, +60 in years 1 and 2 at 10%: NPV = −100 + 54.55 + 49.59 = +4.13; IRR ≈ 13.1%.
BUSINESS EXAMPLE	A forklift costing $45,000 saves $14,000 a year (after tax) for 4 years; residual $5,000 in year 4; required return 10%.
WORKED SOLUTION	PV of savings = 14,000 × 3.16987 = 44,378; PV residual = 5,000 × 0.68301 = 3,415 → NPV = 44,378 + 3,415 − 45,000 = **+$2,793**. IRR ≈ **12.7%**.
COMMON MISTAKES	Discounting the time-0 flow; ignoring residual value or tax; using Excel's NPV on the whole range.

Holding-period return, annualised return and MOIC

FORMULA	**HPR = (End value − Start value + Income) ÷ Start value; Annualised = (1 + HPR)^(12/months) − 1; MOIC = Total value returned ÷ Capital invested**
PLAIN ENGLISH	Total return includes income (dividends, coupons); annualising allows comparison across holding periods. MOIC ignores time; IRR does not.
بالعربية	العائد خلال فترة الاحتفاظ = (القيمة النهائية − القيمة الأولية + الدخل) ÷ القيمة الأولية. ولمقارنة استثمارات بآجال مختلفة نحوّل العائد إلى معدل سنوي. ومضاعف رأس المال (MOIC) لا يأخذ الزمن في الحسبان بخلاف معدل العائد الداخلي.
SIMPLE EXAMPLE	Buy at 100, sell at 110 after 1 year, dividend 3 → 13%.
BUSINESS EXAMPLE	Shares bought at $40, a $1.60 dividend received, sold at $46 after 18 months. Compute HPR and annualised return. Separately: a private investment returns 2.0× in 3 years vs 2.5× in 6 years — compare.
WORKED SOLUTION	HPR = (46 − 40 + 1.6) ÷ 40 = **19.0%**; annualised = 1.19^(12/18) − 1 = **12.3%**. 2.0× in 3 years → IRR = 2^(1/3) − 1 = **26.0%**; 2.5× in 6 years → 2.5^(1/6) − 1 = **16.5%**. Higher MOIC, lower IRR.
COMMON MISTAKES	Ignoring income or costs; comparing returns over different periods without annualising; confusing MOIC with IRR.

Valuation from multiples

FORMULA	**Implied EV = Multiple × Metric (EV/EBITDA, EV/Sales, EV/EBIT); Implied equity = EV − Net debt (and other adjustments); Implied equity (P/E) = P/E × Net income**
PLAIN ENGLISH	Multiples apply the market's pricing of comparable companies to the target's metrics. EV multiples pair with pre-interest metrics; equity multiples with post-interest metrics.
بالعربية	التقييم بالمضاعفات يطبّق تسعير السوق لشركات مماثلة على مؤشرات الشركة المستهدفة. مضاعفات قيمة المنشأة تُطبّق على مؤشرات قبل الفوائد، ومضاعفات حقوق الملكية على مؤشرات بعد الفوائد.
SIMPLE EXAMPLE	EBITDA 10 × 7 = EV 70; net debt 20 → equity 50.
BUSINESS EXAMPLE	Target: EBITDA $14m; net income $6m; net debt $35m; 5m shares. Peers: EV/EBITDA 7.5×, P/E 12×. Implied value per share by each method?
WORKED SOLUTION	EV/EBITDA: EV = 105m → equity 70m → **$14.00/share**. P/E: 12 × 6 = 72m → **$14.40/share**. Consistent within 3%.
COMMON MISTAKES	Applying EV multiples to net income; using forward multiples on trailing metrics; forgetting dilution from options.

43.3 Probability, sensitivity and forecasting

Expected value

FORMULA	**E[X] = Σ p_i × x_i (probabilities sum to 1). Expected loss on credit = PD × LGD × EAD.**
PLAIN ENGLISH	The probability-weighted average outcome. Use it to compare risky choices, but also look at the downside: an option with good expected value may still be unacceptable if its worst case is ruinous.
بالعربية	القيمة المتوقعة هي متوسط النتائج المرجّح باحتمالاتها. تُستخدم للمقارنة بين الخيارات الخطرة، لكن يجب أيضاً النظر إلى أسوأ الاحتمالات، فقد يكون خيار ذو قيمة متوقعة جيدة غير مقبول إذا كانت خسارته المحتملة مدمّرة.
SIMPLE EXAMPLE	60% chance of +100, 40% chance of −50 → 60 − 20 = +40.
BUSINESS EXAMPLE	Selling $100,000 on 60-day open account earns a $12,000 margin. Probability of default 4%; loss given default 60% of the receivable. Credit insurance costs 0.5% of the invoice. Expected value with and without insurance (ignore deductibles)?
WORKED SOLUTION	Without insurance: paid (96%) → +12,000; default (4%) → 12,000 − 60,000 = −48,000. EV = 11,520 − 1,920 = **+$9,600**. With insurance (90% indemnity assumed on the loss): default outcome = 12,000 − 500 − 6,000 = 5,500; paid = 11,500. EV = 0.96 × 11,500 + 0.04 × 5,500 = **+$11,260**. Insurance increases expected value here and caps the downside.
COMMON MISTAKES	Using point estimates as certainties; ignoring correlation (many customers defaulting together in a downturn); forgetting the cost of the risk-reduction tool.

Sensitivity of profit to a driver

FORMULA	**Δ Profit = Δ Driver × Sensitivity; Profit elasticity = %Δ Profit ÷ %Δ Driver**
PLAIN ENGLISH	Measures how much the result moves when one input changes. High sensitivities identify where to focus negotiation, hedging and monitoring.
بالعربية	تحليل الحساسية يقيس مقدار تغير النتيجة عند تغير أحد المدخلات. الحساسيات العالية تحدد أين يجب التركيز في التفاوض والتحوط والمتابعة.
SIMPLE EXAMPLE	If 1 cent on the fuel price per litre changes costs by $5,000, a 20-cent rise costs $100,000.
BUSINESS EXAMPLE	A distributor: revenue $40m; COGS 75% (60% of it priced in EUR); other costs $7m; profit $3m. Sensitivity of profit to a 5% EUR appreciation (no price pass-through)?
WORKED SOLUTION	EUR-linked COGS = 0.75 × 40 × 0.60 = $18m → +5% = **+$0.9m cost** → profit falls to $2.1m (**−30%**). Profit elasticity to EUR = −30% ÷ 5% = **−6**.
COMMON MISTAKES	Changing several inputs at once and calling it sensitivity; ignoring pass-through and hedges already in place.

Simple forecasting: moving average, trend and error

FORMULA	**n-period moving average = (Sum of last n values) ÷ n; Linear trend: $y = a + b \times t$, with $b = \Sigma(t - \bar{t})(y - \bar{y}) \div \Sigma(t - \bar{t})^2$ and $a = \bar{y} - b \times \bar{t}$; MAPE = average of $\lvert$Actual − Forecast$\rvert$ ÷ Actual**
PLAIN ENGLISH	Basic statistical forecasts provide a baseline that judgement and drivers then adjust. Measure forecast accuracy to know how much to trust it.
بالعربية	التنبؤات الإحصائية البسيطة كالمتوسط المتحرك وخط الاتجاه توفر خط أساس يُعدَّل لاحقاً بالحكم المهني ومعرفة المحركات. ويُقاس متوسط الخطأ المطلق النسبي (MAPE) لمعرفة مدى موثوقية التنبؤ.
SIMPLE EXAMPLE	Last three months: 100, 110, 120 → 3-month moving average forecast = 110 (lags a rising trend).
BUSINESS EXAMPLE	Monthly sales (units): 120, 132, 128, 140, 150, 146. Forecast month 7 by 3-month moving average and by linear trend. If actual month 7 is 158, compute each forecast's percentage error.
WORKED SOLUTION	MA = (140 + 150 + 146) ÷ 3 = **145.3**. Trend: $\bar{t} = 3.5$, $\bar{y} = 136$; $\Sigma(t - \bar{t})(y - \bar{y}) = 98$; $\Sigma(t - \bar{t})^2 = 17.5$ → b = **5.6**; a = 136 − 5.6 × 3.5 = 116.4 → forecast = 116.4 + 5.6 × 7 = **155.6**. Errors: MA $\lvert$158 − 145.3$\rvert$ ÷ 158 = **8.0%**; trend **1.5%**.
COMMON MISTAKES	Using moving averages for trending or seasonal data without adjustment; extrapolating trends far ahead; never measuring forecast accuracy.

PRACTICE · Practice questions

1. FOB value $50,000; freight $3,500; insurance 0.15% on 110% of (FOB + freight); duty 10% on CIF; clearance $800. Landed cost?
2. An investment of $200,000 returns $260,000 after 2.5 years with no interim income. Annualised return?
3. Three outcomes for a tender: win at full margin $400k (30%), win with reduced scope $150k (20%), lose −$60k bid costs (50%). Expected value?
4. Peers trade at 1.8× EV/Sales; target sales $25m; net debt $12m. Implied equity value?

ANSWER · Answers to the practice questions

1. C&F 53,500; insurance = 0.0015 × 1.1 × 53,500 = 88.28; CIF = 53,588.28; duty = 5,358.83; landed = 53,588.28 + 5,358.83 + 800 = **$59,747.11**.
2. 1.3^(1/2.5) − 1 = 1.3^0.4 − 1 ≈ **11.1%**.
3. 0.3 × 400 + 0.2 × 150 + 0.5 × (−60) = 120 + 30 − 30 = **+$120k**.
4. EV = 45m → equity = **$33m**.

SUMMARY · Part XX summary

- Know which base a percentage uses (price vs cost; old vs new value; gross vs net of tax).
- Discounts compound; commissions and terms change effective prices.
- Interest depends on rate, day count and compounding; flat rates hide high effective costs.
- In FX, identify the base currency, use bid/ask correctly, and remember forwards reflect interest differentials, not forecasts.
- Convert balance sheets into days to understand cash tied up in operations.
- Landed cost, not supplier price, is the basis for pricing imports.
- Expected values support decisions only together with an understanding of the downside.

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